

Gravitational waves and a spherical thin shell: linearized junction conditions, scattering coefficients, quasinormal modes and the dynamical fate of the background

A complete account of the first-round work (“First round” folder),
written and verified in the second round

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Abstract

We give a self-contained account of a computation of gravitational-wave scattering off, and of the quasinormal modes (QNMs) of, a spherical thin shell that separates a flat interior from a Schwarzschild exterior. Both parities are treated: odd (Regge–Wheeler, Cunningham–Price–Moncrief) and even (Zerilli–Moncrief). The first obstacle is physical. A pure domain wall, the object of the prior notes, has no static configuration: it always collapses. We examine every way out. A static dust shell does not exist. A static perfect-fluid shell exists, but its pressure-to-density ratio is fixed by the compactness (model C). A domain wall can be frozen at its turning point (model A). The linearized Israel junction conditions are then derived from first principles and verified symbolically. In the odd sector they reduce, for every perfect-fluid shell, to continuity of the master function and a repulsive δ -function potential of strength $\Delta_{\text{odd}} = \sqrt{f_R}(1 - \sqrt{f_R})/R$ in the tortoise coordinate. In the even sector they form a frequency-dependent 2×2 transfer matrix that encodes the radial displacement of the shell in the two Regge–Wheeler gauges and the motion of its matter. Reflection and transmission coefficients are defined by opening the interior, and satisfy $\mathcal{R} + \mathcal{T} = 1$ to 10^{-9} . They tend to one at low frequency, fall as $\Delta^2/4\omega^2$ at high frequency, and approach the black-hole reflectivity as $R \rightarrow 2M$. Two independent numerical methods agree. The QNM spectrum contains wave modes, trapped modes of ultracompact shells and four even-parity matter modes per multipole, one of which is purely imaginary and growing. This reproduces the recent result of Pitre, Schneider and Poisson that the static shell is dynamically unstable. A second, independent verification round confirms the odd junction through a thick-shell limit, confirms the matter modes through a Newtonian derivation, and shows that the even-parity results of the frozen domain wall are not a controlled approximation.

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1 Introduction

Black holes are the simplest compact objects of general relativity. Whether astrophysical dark compact objects really possess event horizons can be probed through the way they respond to gravitational perturbations [40–43]. The response of a Schwarzschild black hole is well understood: the Regge–Wheeler (RW) [1] and Zerilli [2] equations describe the odd- and even-parity perturbations. The black hole reflects low-frequency waves and absorbs high-frequency ones [3], and it rings down in a discrete set of quasinormal modes [15, 19, 44–46]. Horizonless objects such as gravastars [11, 37–39] differ in two ways. They have a surface, where the perturbation must satisfy junction conditions, and they have an interior, where waves can be trapped.

The simplest such object is a *thin spherical shell* of matter at areal radius $R > 2M$. Birkhoff’s theorem and regularity at the centre force the interior to be flat [9], and the exterior is Schwarzschild. The shell is described by a surface stress-energy tensor S_{ab} on a timelike hypersurface Σ . The Lanczos–Israel formalism [5, 6, 35, 36] relates the jump of the extrinsic curvature across Σ to S_{ab} . The task considered here is to compute, for both parities and several multipoles ℓ and shell parameters: (i) the reflection and transmission coefficients $\mathcal{R}(\omega)$, $\mathcal{T}(\omega)$ of gravitational waves; and (ii) the QNM spectrum. Frequencies are in units of M^{-1} , $G = c = 1$, and the time dependence is $e^{-i\omega t}$.

The prior notes [9, 10] had established the interior solution and set up the Israel formalism for a domain wall. Along the way they discovered a serious obstacle: a spherical domain wall with $M \neq 0$ cannot be static [5]. This paper first explains how the obstacle is handled (Sec. 2). It then develops the perturbation theory (Secs. 3–4) and derives the junction conditions (Sec. 5). Finally it defines the scattering problem (Sec. 6), describes the numerical methods (Sec. 7) and presents and discusses the results (Secs. 8–9).

A literature search showed that the static perfect-fluid shell (our model C) was studied very recently by Pitre, Schneider and Poisson [12] [PSP26]. They computed its QNMs with a different formalism and found it dynamically unstable on all angular scales, complementing an eikonal analysis by Yang, Bonga and Pan [14]. We reproduce their results and use them as validation. The reflection and transmission coefficients, the explicit δ -function form of the odd junction, and the adiabatic domain-wall model appear to be new.

Boxed remarks labelled *Second-round verification* summarise independent checks. They mark the few places where the verification changed a conclusion of the first round. The full evidence is in the companion verification report.

2 The background: thin shells and the collapse problem

2.1 Thin-shell formalism

Let Σ be the world-tube of the shell, with intrinsic coordinates $y^a = (\tau, \theta, \phi)$, tangent vectors $e_a^\alpha = \partial x^\alpha / \partial y^a$, and unit normal n_α pointing from the interior (–) to the exterior (+). The induced metric and the extrinsic curvature are [5, 6]

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad K_{ab} = e_a^\alpha e_b^\beta \nabla_\alpha n_\beta = -n_\mu (\partial_a e_b^\mu + \Gamma^\mu_{\nu\rho} e_a^\nu e_b^\rho). \quad (2.1)$$

The junction conditions are the continuity of the induced metric (Poisson’s Eq. 3.49) and the Lanczos–Israel condition (Poisson 3.56; IS84 2.6),

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab}, \quad [X] \equiv X|_+ - X|_-. \quad (2.2)$$

When both sides are vacuum, the Codazzi equation turns Eq. (2.2) into the conservation law $D_b S^{ab} = 0$ (IS84 2.8a). We take a perfect-fluid surface tensor, $S^{ab} = \sigma u^a u^b + p(h^{ab} + u^a u^b)$, with surface energy density σ and surface pressure p . Ipser and Sikivie write it with the tension $\tau = -p$ (IS84 2.9). A domain wall has $\tau = \sigma$, i.e. $S_{ab} = -\sigma h_{ab}$ (IS84 2.10), and conservation then makes σ a constant (IS84 2.12).

2.2 Spherical shell

Inside, $ds_-^2 = -dT^2 + dr^2 + r^2 d\Omega^2$. Outside, $ds_+^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - 2M/r$. For a shell $r = R(\tau)$ define $\alpha = (1 + \dot{R}^2)^{1/2}$ and $\beta = (f(R) + \dot{R}^2)^{1/2}$, where a dot denotes $d/d\tau$ (IS84 3.4). The nonzero mixed components of K are (IS84 3.5–3.6; Poisson 3.65)

$$K^\theta_\theta|_+ = \frac{\beta}{R}, \quad K^\theta_\theta|_- = \frac{\alpha}{R}, \quad K^\tau_\tau|_+ = \frac{\ddot{R} + M/R^2}{\beta}, \quad K^\tau_\tau|_- = \frac{\ddot{R}}{\alpha}. \quad (2.3)$$

The mixed Israel equations $[K^a_b] - \delta^a_b [K] = -8\pi S^a_b$ with $S^a_b = \text{diag}(-\sigma, p, p)$ give

$$\sigma = -\frac{[K^\theta_\theta]}{4\pi} = \frac{\alpha - \beta}{4\pi R}, \quad p = \frac{[K^\tau_\tau] + [K^\theta_\theta]}{8\pi}. \quad (2.4)$$

These are equivalent to IS84's equations of motion (3.7a,b),

$$(\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} - \frac{2\tau}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}, \quad (\alpha - \beta)\ddot{R} = -\frac{\alpha M}{R^2} + 4\pi(\sigma - 2\tau)\alpha\beta. \quad (2.5)$$

Using $\alpha^2 - \beta^2 = 2M/R$, the first of Eqs. (2.4) becomes the first integral $M = \frac{1}{2}(\alpha + \beta)4\pi\sigma R^2$ (IS84 3.8). At an instant with $\dot{R} = 0$ this reads

$$4\pi\sigma R = 1 - \sqrt{f_R} \quad \Longleftrightarrow \quad M = 4\pi\sigma R^2(1 - 2\pi\sigma R), \quad (2.6)$$

which is IS84 Eq. (3.9). The same relation holds at the turning point of a dust shell, as follows from Poisson's Eq. (3.70), $M = m(1 + \dot{R}^2)^{1/2} - m^2/2R$ with conserved $m = 4\pi\sigma R^2$.

2.3 The collapse problem and the four options

Why a domain wall cannot sit still. For $\tau \geq 0$ the right-hand side of the first of Eqs. (2.5) is negative. As Ipser and Sikivie put it, “ $\ddot{R}$ is always negative if $\tau \geq 0$. Hence, a spherical wall with $\tau \geq 0$ always collapses to a black hole, regardless of its size” [5]. A monochromatic scattering calculation presupposes a time-independent background, so this must be resolved first. The prior notes [10] listed three ways out and the task statement four (A–D).

Static shells need negative tension. Setting $\dot{R} = \ddot{R} = 0$ in Eq. (2.5) gives $M/R^2 + (2\tau/\sigma)(f_R + \sqrt{f_R})/R = 0$, i.e.

$$\frac{\tau}{\sigma} = -\kappa, \quad \kappa = \frac{M}{2R\sqrt{f_R}(1 + \sqrt{f_R})} = \frac{1 - \sqrt{f_R}}{4\sqrt{f_R}} > 0. \quad (2.7)$$

A static shell needs a *positive surface pressure* $p = \kappa\sigma$. Explicitly (Poisson 3.80–3.81),

$$\sigma = \frac{1 - \sqrt{f_R}}{4\pi R}, \quad p = \frac{1 - M/R - \sqrt{f_R}}{8\pi R\sqrt{f_R}} = \frac{(1 - \sqrt{f_R})^2}{16\pi R\sqrt{f_R}}. \quad (2.8)$$

Option B (static dust) does not exist. For dust $p = 0$, and Eq. (2.5) gives $(\alpha + \beta)\ddot{R} = -\alpha M/R^2 < 0$. A dust shell is at most momentarily static, at its turning point, where it has the same geometry (2.6) as the wall and $\ddot{R} = -M/[R^2(1 + \sqrt{f_R})]$. The task statement suggested $M = 4\pi\sigma R^2\sqrt{f_R}$ for a static dust shell. That relation is the pressure balance $[K^\tau_\tau] = 4\pi\sigma$ with $p = 0$, and it contradicts the angular condition $4\pi\sigma R = 1 - \sqrt{f_R}$ unless $M = 0$.

Option C (static perfect fluid): chosen as primary, model C. Equation (2.7) shows that the ratio κ is *not* a free parameter: it is fixed by the compactness. The task statement had proposed studying the scattering as a function of κ . The family is therefore parametrized by R/M (Table 1). The dominant energy condition $p \leq \sigma$ means $\kappa \leq 1$, i.e. $\sqrt{f_R} \geq 1/5$ or $R \geq 25M/12 \simeq 2.083M$. Perturbations also require an equation of state for the *response* of the shell, $\delta p = v_s^2 \delta\sigma$. Following

Table 1: Background quantities. $(M/R^3)^{1/2}$ sets the scale of the model-C matter modes and instability; ω_{dyn} is the wall’s collapse rate.

R/M	σM	pM	$\kappa = p/\sigma$	$v_s^2(\Gamma=2)$	I&S (3.7a) resid.	$\ddot{R}_{\text{wall}}M$	$\omega_{\text{dyn}}M$	$(M/R^3)^{1/2}M$
2.2	0.02527	1.463e-02	0.5792	0.7335	0.00e+00	-0.4328	0.1337	0.3065
3.0	0.01121	2.052e-03	0.1830	0.3094	1.39e-17	-0.4553	0.2249	0.1925
6.0	0.00243	1.367e-04	0.0562	0.1064	0.00e+00	-0.2875	0.1787	0.0680
10.0	0.00084	2.479e-05	0.0295	0.0573	1.73e-18	-0.1842	0.1214	0.0316

PSP26 we take a barotropic shell with adiabatic index $\Gamma = (\sigma/p) dp/d\sigma$ (their Eqs. 2.4–2.7, where their energy density μ is our σ). This gives

$$\delta p = \Gamma \frac{p}{\sigma + p} \delta \sigma, \quad v_s^2 = \frac{\Gamma \kappa}{1 + \kappa}, \quad (2.9)$$

with fiducial value $\Gamma = 2$. Causality ($v_s^2 \leq 1$) holds for $\Gamma = 2$ exactly when $\kappa \leq 1$, the same bound. Results are therefore given for $R \geq 2.1M$ except when the black-hole limit is explored.

Option A (domain wall frozen at its turning point): model A. For $\tau = \sigma$ and $\dot{R} = 0$, Eqs. (2.3)–(2.4) give $[K^\tau_\tau] = [K^\theta_\theta] = -4\pi\sigma$, hence

$$\ddot{R} = -\frac{M}{R^2(1 + \sqrt{f_R})} - \frac{2\sqrt{f_R}}{R} = -\frac{1 + 3\sqrt{f_R}}{2R}. \quad (2.10)$$

The first form is IS84 (3.7a) with $\alpha = 1$, $\beta = \sqrt{f_R}$. The frozen-background approximation requires frequencies well above the dynamical rate, defined in exterior time as $\omega_{\text{dyn}} = \sqrt{f_R}(|\ddot{R}|/R)^{1/2}$.

Option D (Schwarzschild/de Sitter interior). Not implemented. Its de Sitter version, with a surface-tension-only shell, is the thin-shell gravastar of Pani et al. [11]. Model C is the flat-interior, $\sigma \neq 0$ member of the same family.

Second-round verification. All the relations of this section were re-derived symbolically, and Table 1 was reproduced to every printed digit. One statement of the first round needs correcting. It said that “every thin shell of this type evolves on the timescale $(R^3/M)^{1/2}$ ”. For large R , Eq. (2.10) gives $\omega_{\text{dyn}} \rightarrow \sqrt{2}/R$: the tension-driven wall collapses on the light-crossing time, whatever M is. This rate is comparable to the wave-mode frequencies ($\sim 1/R$), as Fig. 1 shows. The adiabatic condition $\omega \gg \omega_{\text{dyn}}$ is therefore only marginally satisfied by the wall’s QNMs. By contrast, the model-C instability rate $\simeq 0.6(M/R^3)^{1/2}$ is well separated from them.

2.4 Time normalization: the interior blueshift

At a static shell, continuity of h_{ab} requires $dT = \sqrt{f_R} dt$ (Poisson’s interior metric 3.75). A perturbation $\propto e^{-i\omega t}$ outside is therefore $\propto e^{-i\nu T}$ inside, with

$$\nu = \omega/\sqrt{f_R}. \quad (2.11)$$

The regular interior solution is $\hat{j}_\ell(\nu r)$, not $\hat{j}_\ell(\omega r)$ as in the prior notes and the task statement. PSP26 use the same normalization, $w = F^{-1/2}\omega$. It is convenient to introduce a global tortoise coordinate

$$x = r_* = r + 2M \ln(r/2M - 1) \quad (r > R), \quad x = r_*(R) + (r - R)/\sqrt{f_R} \quad (r < R), \quad (2.12)$$

in which both regions obey $-\partial_t^2 \Psi + \partial_x^2 \Psi - V \Psi = 0$, with $V_{\text{int}} = f_R \ell(\ell + 1)/r^2$ (Fig. 2).

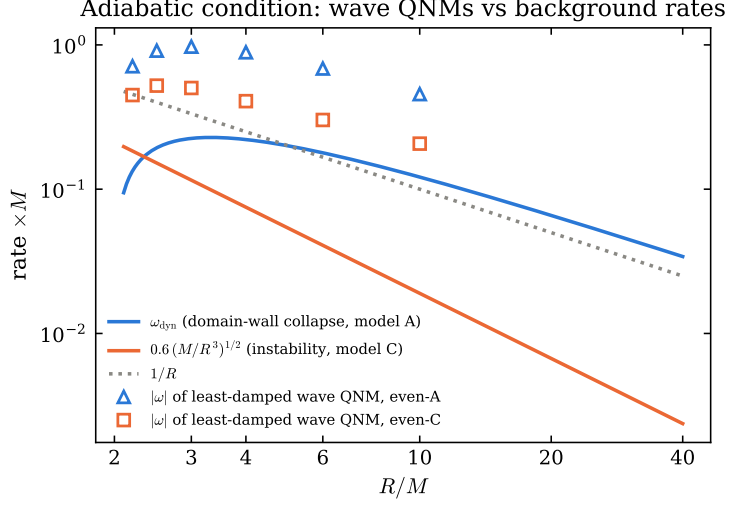


Figure 1: Background rates versus the least-damped wave QNMs ($\ell = 2$; second-round figure).

3 Perturbation equations

We follow Martel and Poisson [4] [MP05]. The odd harmonics are $X_A = -\varepsilon_A^B D_B Y$ and $X_{AB} = -\frac{1}{2}(\varepsilon_A^C D_B + \varepsilon_B^C D_A) D_C Y$. The even ones are $Y_A = D_A Y$ and $Y_{AB} = [D_A D_B + \frac{1}{2}\ell(\ell+1)\Omega_{AB}]Y$ (MP05 3.1–3.7). We write $\lambda = \ell(\ell+1)$ and $\mu = (\ell-1)(\ell+2)$, and we use Regge–Wheeler gauge *on each side separately*: $h_2 = 0$ in the odd sector, $j_a = G = 0$ in the even sector [1, 4].

3.1 Odd parity

With $p_{aB} = h_a X_B$, $h_a = (h_t, h_r)$ (MP05 5.2), the Cunningham–Price–Moncrief function [33]

$$\Psi_{\text{odd}} = \frac{2r}{\mu} \left(\partial_r h_t - \partial_t h_r - \frac{2}{r} h_t \right) \quad (3.1)$$

(MP05 5.13) obeys the Regge–Wheeler equation (MP05 5.14–5.15; RW57 Eq. 25)

$$\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V_{\text{RW}}] \Psi = 0, \quad V_{\text{RW}} = f \left[\frac{\lambda}{r^2} - \frac{6M}{r^3} \right]. \quad (3.2)$$

In vacuum the field equations can be inverted:

$$h_t = \frac{f}{2} \partial_r (r \Psi_{\text{odd}}), \quad h_r = \frac{r}{2f} \partial_t \Psi_{\text{odd}} = -\frac{i\omega r}{2f} \Psi_{\text{odd}}. \quad (3.3)$$

The second relation is MP05’s (5.18), $\Psi_{\text{RW}} = \frac{1}{2} \partial_t \Psi_{\text{odd}}$ in vacuum. The first follows from the gauge-invariant equation $\nabla_a h^a = 0$ (MP05 5.9).

3.2 Even parity

With $h_{tt} = f H_0$, $h_{tr} = H_1$, $h_{rr} = H_2/f$ and $p_{AB} = r^2 K \Omega_{AB} Y$, the trace-free angular equation gives $H_0 = H_2 \equiv H$. The Zerilli–Moncrief function [4, 34],

$$\Psi_{\text{even}} = \frac{2r}{\lambda} \left[K + \frac{2f}{\Lambda} (H - rK') \right], \quad \Lambda = \mu + \frac{6M}{r}, \quad (3.4)$$

obeys Eq. (3.2) with the Zerilli potential (Zerilli Eq. 5; $f \times$ MP05 4.26)

$$V_Z = f \frac{2n^2(n+1)r^3 + 6n^2 M r^2 + 18n M^2 r + 18M^3}{r^3(nr + 3M)^2}, \quad n = \mu/2. \quad (3.5)$$

Effective potentials: exterior V_{RW}, V_Z ; flat interior $f_R \ell(\ell+1)/r^2$; shell = repulsive δ -coupling (odd: $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$)

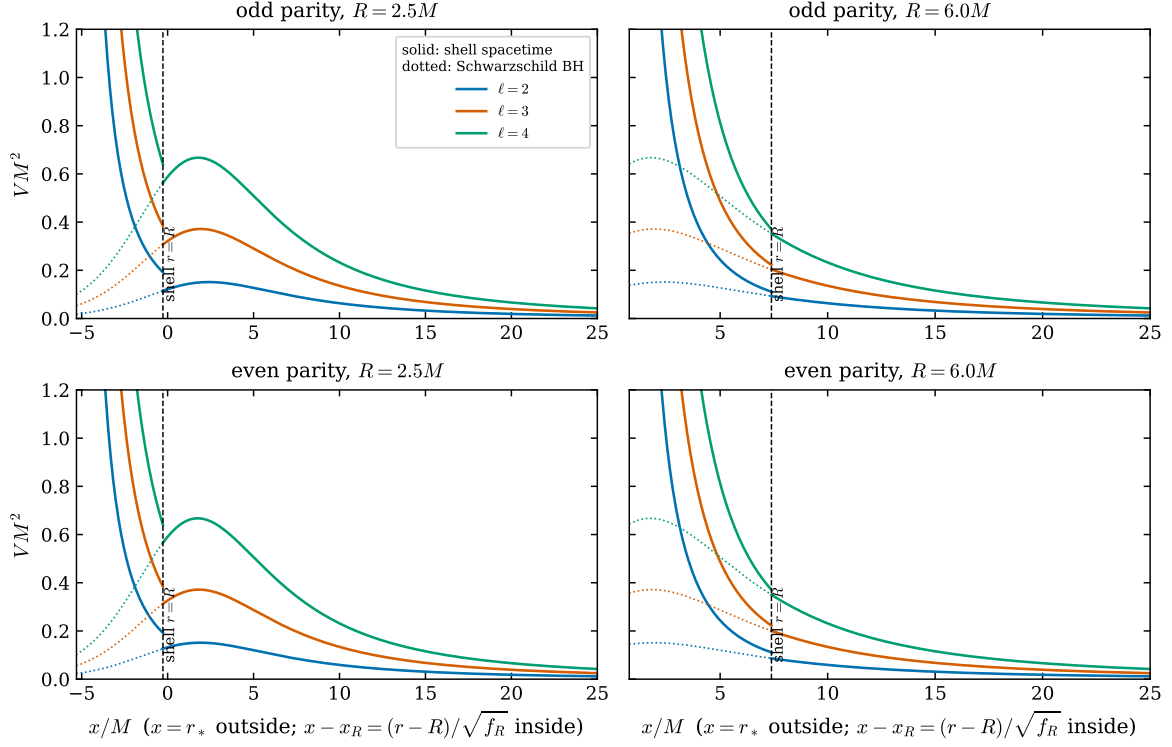


Figure 2: Effective potentials in the global tortoise coordinate for $\ell = 2, 3, 4$ (first-round figure). Exterior: V_{RW} and V_Z . Interior: $f_R \ell(\ell+1)/r^2$. Dotted curves: the Schwarzschild continuation.

The linearized Einstein equations give a first-order system for (K, H, H_1) plus an algebraic identity (Zerilli 1970, p. 737). Inverting Eq. (3.4) with them gives the reconstruction

$$K = f \Psi' + \frac{\mu(\mu+2)r^2 + 6\mu M r + 24M^2}{2r^2(\mu r + 6M)} \Psi, \quad (3.6)$$

$$H_1 = -i\omega \left[r \Psi' + \frac{\mu r^2 - 3\mu M r - 6M^2}{(r-2M)(\mu r + 6M)} \Psi \right], \quad (3.7)$$

$$H = f \frac{\mu r^2 - 3\mu M r - 6M^2}{(r-2M)(\mu r + 6M)} \Psi' + \left[\frac{Q(r)}{2r^2(r-2M)(\mu r + 6M)^2} - \frac{\omega^2 r}{f} \right] \Psi, \quad (3.8)$$

$$Q(r) = \mu^2(\mu+2)r^4 - 2M\mu^2(\mu-1)r^3 - 12M^2\mu(\mu-3)r^2 - 72M^3(\mu-1)r - 144M^4.$$

How these relations were established. Nothing in this section was copied from the literature. It was re-derived by brute force. The symbolic code linearizes the Einstein tensor for a generic axisymmetric harmonic with λ symbolic, reduces the angular dependence with the Legendre equation, and eliminates Ψ'' with the master equation. The residuals of all six even-parity field equations vanish identically in exact rational arithmetic at random rational points, and so do the five odd-parity ones. Equation (3.4) returns Ψ , and V_Z equals f times MP05's (4.26).

3.3 Chandrasekhar transformation

The map [11, 29]

$$Z = \left[\mu(\mu+2) + \frac{72M^2 f}{r(\mu r + 6M)} \right] \Psi_{\text{RW}} + 12M \frac{d\Psi_{\text{RW}}}{dr_*} \quad (3.9)$$

takes solutions of the RW equation into solutions of the Zerilli equation for every ω (verified symbolically). Since $d\Psi/dr_* = i\omega\Psi$ for outgoing waves, it maps $e^{i\omega r_*}$ into $[\mu(\mu+2) + 12iM\omega]e^{i\omega r_*}$.

It is used to build Zerilli solutions from RW ones, and it is the reason the two potentials share the Schwarzschild spectrum (isospectrality).

4 Interior solution

With $M = 0$ both potentials reduce to λ/r^2 [9], so inside

$$\Psi'' + [\nu^2 - \lambda/r^2]\Psi = 0 \quad (0 < r < R), \quad (4.1)$$

solved by the Riccati–Bessel functions $\hat{j}_\ell(z) = zj_\ell(z)$ and $\hat{y}_\ell(z) = zy_\ell(z)$ with $z = \nu r$ [7–9]. Regularity at the centre selects

$$\Psi_{\text{int}} = A\hat{j}_\ell(\nu r), \quad \hat{j}_\ell(z) \simeq \frac{z^{\ell+1}}{(2\ell+1)!!} \quad (z \rightarrow 0). \quad (4.2)$$

Both parities have the same interior profile and differ only through the junction conditions. For the open-interior scattering problem of Sec. 6 we also need the Riccati–Hankel functions $\hat{h}_\ell^{\text{out/in}} = -\hat{y}_\ell \pm i\hat{j}_\ell \sim e^{\pm i(z-\ell\pi/2)}$. With the Wronskian $\hat{j}_\ell\hat{y}'_\ell - \hat{j}'_\ell\hat{y}_\ell = 1$ (DLMF 10.50.1) one has $\text{Im}(\hat{h}^{\text{in}*}\hat{h}^{\text{in}'}) = -1$. The reconstruction formulas of Sec. 3 hold inside with $M = 0$ and $\omega \rightarrow \nu$, because the interior is written in its own time T .

5 Linearized junction conditions

5.1 Method

On each side the bulk perturbation is written in that side’s own RW gauge. RW gauge fixes the bulk gauge freedom completely for $\ell \geq 2$, but it is not adapted to the shell: in the two gauges the shell sits at different perturbed positions. The world-tube is therefore described by a perturbed embedding on each side (cf. Pani et al. [11], Sec. III.C and App. A):

$$x_\pm^\mu(y) = X_{0\pm}^\mu(y) + \epsilon Z_\pm^\mu(y), \quad Z_+ = (0, \xi_+ Y, 0, 0), \quad Z_- = (\zeta Y, \xi_- Y, \eta \partial_\theta Y, 0) [+ \chi X^\phi \text{ (odd)}]. \quad (5.1)$$

The intrinsic coordinates are chosen to coincide with the exterior ones. $\xi_\pm$ are the radial displacements of the shell in the two gauges, and ζ, η, χ describe the relative identification of time and angles. The unit normal is solved to first order from $n_\mu e_a^\mu = 0$ and $g^{\mu\nu} n_\mu n_\nu = 1$. The induced metric and Eq. (2.1) are expanded to $O(\epsilon)$, including the displacement of the background fields, $Q(X_0 + \epsilon Z) = Q(X_0) + \epsilon Z^\mu \partial_\mu Q$. The result is projected onto harmonics, and the bulk functions are replaced by master functions through Eqs. (3.3)–(3.8). The shell matter is perturbed as $\sigma \rightarrow \sigma + \delta\sigma Y$, $p \rightarrow p + v_s^2 \delta\sigma Y$, with $u_A = U D_A Y$ (even) or $U X_A$ (odd). For the wall, $\delta S_{ab} = -\sigma \delta h_{ab}$. The junction conditions (2.2) then become a linear system

$$A(\omega) \mathbf{x} = B(\omega) \begin{pmatrix} \Psi_-(R) \\ \Psi'_-(R) \end{pmatrix}, \quad \mathbf{x} = (\text{shell variables}, \Psi_+(R), \Psi'_+(R)), \quad (5.2)$$

whose last two components define the transfer matrix, $(\Psi_+, \Psi'_+)^T = \mathbb{T}(\Psi_-, \Psi'_-)^T$.

5.2 Odd parity

Odd perturbations do not move the shell radially. Since g^{rr} is unperturbed, $n_\alpha = f^{-1/2} \partial_\alpha r$ stays exact at first order, so $K_{ab} = -N \Gamma^r_{ab}$ with $N = f^{-1/2}$. The first-order Christoffel symbols are $\delta \Gamma^r_{tA} = \frac{f}{2} (\partial_t h_r - \partial_r h_t) X_A$ (the connection terms of the three covariant derivatives cancel in pairs) and $\delta \Gamma^r_{AB} = f h_r X_{AB}$ (using $D_A X_B + D_B X_A = 2X_{AB}$). Hence

$$\delta K_{tA} = \frac{\sqrt{f}}{2} (\partial_r h_t - \partial_t h_r) X_A, \quad \delta K_{AB} = -\sqrt{f} h_r X_{AB}. \quad (5.3)$$

The first of these is Eq. (12) of the prior notes [10]. By Eq. (3.1), $\partial_r h_t - \partial_t h_r = \mu\Psi/(2r) + 2h_t/r$. In proper time on Σ ($e_\tau = f_R^{-1/2}\partial_t$ outside, ∂_T inside), with the common quantity $\mathcal{H} = \delta h_{\tau A}$,

$$\text{outside: } \mathcal{H} = h_t^+/\sqrt{f_R}, \quad \delta K_{\tau A}^+ = \frac{\mu\Psi_+}{4R} + \frac{\sqrt{f_R}\mathcal{H}}{R}; \quad \text{inside: } \mathcal{H} = h_T^-, \quad \delta K_{\tau A}^- = \frac{\mu\Psi_-}{4R} + \frac{\mathcal{H}}{R}. \quad (5.4)$$

Continuity of δh_{AB} forces $\chi = 0$. Three equations remain.

1. *AB Israel equation.* $\delta S_{AB}^{\text{odd}} = 0$ (since $\delta h_{AB} = 0$ and $u_A u_B = O(\epsilon^2)$), so $[\sqrt{f_R}h_r] = 0$. With $h_r^+ = -i\omega R\Psi_+/2f$ and $h_r^- = -i\nu R\Psi_-/2$, $\nu = \omega/\sqrt{f_R}$, this gives $\Psi_+(R) = \Psi_-(R)$ exactly.
2. *τA Israel equation.* With $\delta S_{\tau A} = -(\sigma + p)U + p\mathcal{H}$ and $[K] = 4\pi(2p - \sigma)$, the left-hand side is $\mu[\Psi]/4R + (\sqrt{f_R} - 1)\mathcal{H}/R$, where $(\sqrt{f_R} - 1)/R = -4\pi\sigma$. This leaves $\mu[\Psi]/4R = 8\pi(\sigma + p)U$, hence $U = 0$: odd perturbations do not set the shell matter in motion, in agreement with $\partial_\tau u_A = 0$ from $D_b S^{ab} = 0$. The background trace $[K]^{(0)}$, flagged as “the remaining step” in [10], combines with the pressure term into $[K]^{(0)} - 8\pi p = -4\pi\sigma$. The dependence on p , and with it on $\ddot{R}$ for the wall, cancels identically.
3. *Continuity of $\mathcal{H}$:* $\frac{1}{2}(r\Psi_-)'|_R = \frac{\sqrt{f_R}}{2}(r\Psi_+)'|_R$.

Together:

$$\boxed{\Psi_+ = \Psi_-, \quad \sqrt{f_R}\Psi'_+ - \Psi'_- = \frac{1 - \sqrt{f_R}}{R}\Psi = 4\pi\sigma\Psi} \quad \Longleftrightarrow \quad \left[\frac{d\Psi}{d\ell_p}\right] = 4\pi\sigma\Psi, \quad (5.5)$$

where ℓ_p is proper radial distance. In the global coordinate (2.12), $[d\Psi/dx] = \Delta_{\text{odd}}\Psi$ with

$$\Delta_{\text{odd}} = \frac{\sqrt{f_R}(1 - \sqrt{f_R})}{R} = 4\pi\sigma\sqrt{f_R} > 0. \quad (5.6)$$

The shell acts as a *repulsive δ -function potential* $\Delta_{\text{odd}}\delta(x - x_R)$, independent of ω and of the equation of state. The transfer matrix is $\mathsf{T}_{\text{odd}} = \begin{pmatrix} 1 & 0 \\ (1 - \sqrt{f_R})/(R\sqrt{f_R}) & 1/\sqrt{f_R} \end{pmatrix}$, with $\det \mathsf{T}_{\text{odd}} = f_R^{-1/2}$. The same condition follows from the brute-force symbolic system, for model C (4 equations, 4 unknowns) and for model A (4 equations, 3 unknowns, consistent with rank 3). It agrees with PSP26 Eq. (5.23), $F^{1/2}\eta_{\text{out}} - \eta_{\text{in}} + F^{1/2} - 1 = 0$ with $\eta = R\psi'/\psi$: the phase factors of their retarded-time slicing cancel because $w = F^{-1/2}\omega$. It also agrees with Pani et al.’s (3.29) and (A35), $[h_0] = 0$ and $[\sqrt{h}h_1] = 0$, and it reduces to the identity (continuity of Ψ and Ψ' , the “inviscid” limit of Boyanov et al. [13]) as $M \rightarrow 0$.

Second-round verification. The second round derived the same result by a completely independent route (companion report and App. B). Linearizing the Einstein equations for axial perturbations of a smooth, static, *anisotropic* fluid gives the master potential $V = e^{2\Phi}[\ell(\ell + 1)/r^2 - 6m/r^3 + 4\pi(\rho - p_r)]$. It generalises the familiar stellar result of Chandrasekhar and Ferrari [30]; see also [31, 32]. For a shell with $p_r = 0$ the $4\pi\rho$ term becomes $\Delta_{\text{odd}}\delta(x - x_R)$, which is exactly Eq. (5.6). A numerical thick-shell computation converges linearly in the thickness to the junction (5.5) and to none of the alternatives tried (Fig. 3).

5.3 Even parity

Induced metric. The first-order induced metric from the two sides is (coefficients of Y , $D_A Y$, $\Omega_{AB}Y$ and Y_{AB})

$$\begin{aligned} \text{outside: } \delta h_{\tau\tau} &= H_+ - \frac{2M}{R(R-2M)}\xi_+, & \delta h_{\tau A} &= 0, \\ \delta h_{AB}^{\text{tr}} &= R^2 K_+ + 2R\xi_+, & \delta h_{AB}^{\text{tf}} &= 0, \\ \text{inside: } \delta h_{\tau\tau} &= H_- + 2i\nu\zeta, & \delta h_{\tau A} &= -\zeta - i\nu R^2\eta, \\ \delta h_{AB}^{\text{tr}} &= R^2 K_- - \lambda R^2\eta + 2R\xi_-, & \delta h_{AB}^{\text{tf}} &= 2R^2\eta. \end{aligned} \quad (5.7)$$

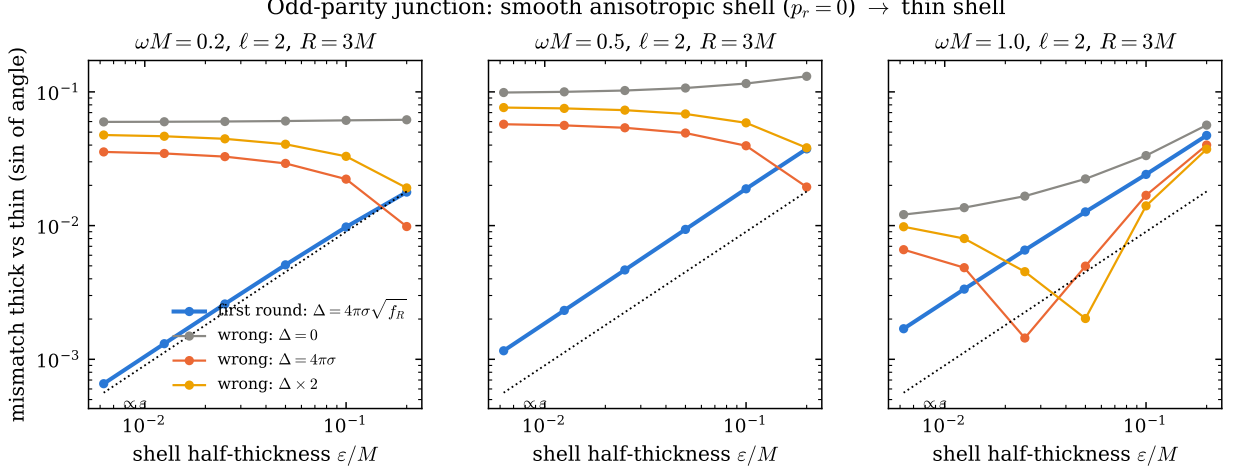


Figure 3: Thick-shell test of the odd junction (second round): mismatch between a smooth anisotropic shell of half-thickness ε and the thin-shell prediction.

The first junction condition forces $\eta = \zeta = 0$ and leaves $H_+ - \frac{2M}{R(R-2M)}\xi_+ = H_-$ and $RK_+ + 2\xi_+ = RK_- + 2\xi_-$, in agreement with Pani et al.'s (A16), (A20). Because $\xi_+ \neq \xi_-$, the shell's radial displacement is different in the two RW gauges. This is why the even sector cannot reduce to a scalar jump condition.

Israel equations. The four even components ($\tau\tau$, τA , trace and trace-free AB) of $[\delta K_{ab} - \delta h_{ab}K - h_{ab}\delta K] = -8\pi\delta S_{ab}$ involve $H_{\pm}, H'_{\pm}, K_{\pm}, K'_{\pm}, H_{1\pm}, \xi_{\pm}$ and the matter variables. For model C,

$$\delta S_{\tau\tau} = \delta\sigma - \sigma\delta h_{\tau\tau}, \quad \delta S_{\tau A} = -(\sigma + p)U + p\delta h_{\tau A}, \quad \delta S_{AB}^{\text{tr}} = v_s^2\delta\sigma R^2 + p\delta h_{AB}^{\text{tr}}, \quad \delta S_{AB}^{\text{tf}} = p\delta h_{AB}^{\text{tf}}. \quad (5.8)$$

Model C gives 8 equations for the 8 unknowns $(\xi_+, \xi_-, \zeta, \eta, \delta\sigma, U, \Psi_+, \Psi'_+)$, with full rank. Its solution defines $\mathsf{T}_{\text{even}}^{\text{C}}(\omega; \ell, R/M, \Gamma)$.

Structure of the result.

- Ψ is *discontinuous*. T_{even} is a full 2×2 matrix that depends on frequency (through the ω^2 term of H and through the shell's inertia) and, through v_s^2 , on the shell's equation of state. The even problem is a genuine 2×2 matching, as the task statement anticipated might happen.
- $\det \mathsf{T}_{\text{even}}^{\text{C}} = f_R^{-1/2}$ exactly, and T is real for real ω , so the flux is conserved: this is the identity that makes $\mathcal{R} + \mathcal{T} = 1$ (Sec. 6). It is *not* built into the construction, so it is a strong test.
- $\mathsf{T}_{\text{even}} \rightarrow \mathbb{I}$ as $M \rightarrow 0$ at fixed R (linearly in M).
- The matrix has poles on the matter scale $\omega \sim (M/R^3)^{1/2}$, which the odd sector lacks. They signal the shell's own oscillations.

High-precision arithmetic. The exact matrices involve large cancellations near $R = 2M$. In double precision $\det \mathsf{T}\sqrt{f_R} - 1$ is $\sim 10^{-7}$ at $2.2M$ and the method fails below $\simeq 2.05M$. For $R < 2.3M$ the system is therefore solved with 40-digit `mpmath`, and $\det \mathsf{T}\sqrt{f_R} = 1$ then holds to 10^{-15} down to $2.01M$.

Second-round verification. The second round verified $\det \mathsf{T}\sqrt{f_R} = 1$ to 1.1×10^{-12} over 300 random real and complex parameter sets, the reality of T , flux conservation to 10^{-13} , and linear convergence to $\mathbb{I}$ as

Model A is prescription dependent: which 2 of the 4 Israel equations are imposed (metric continuity always imposed)

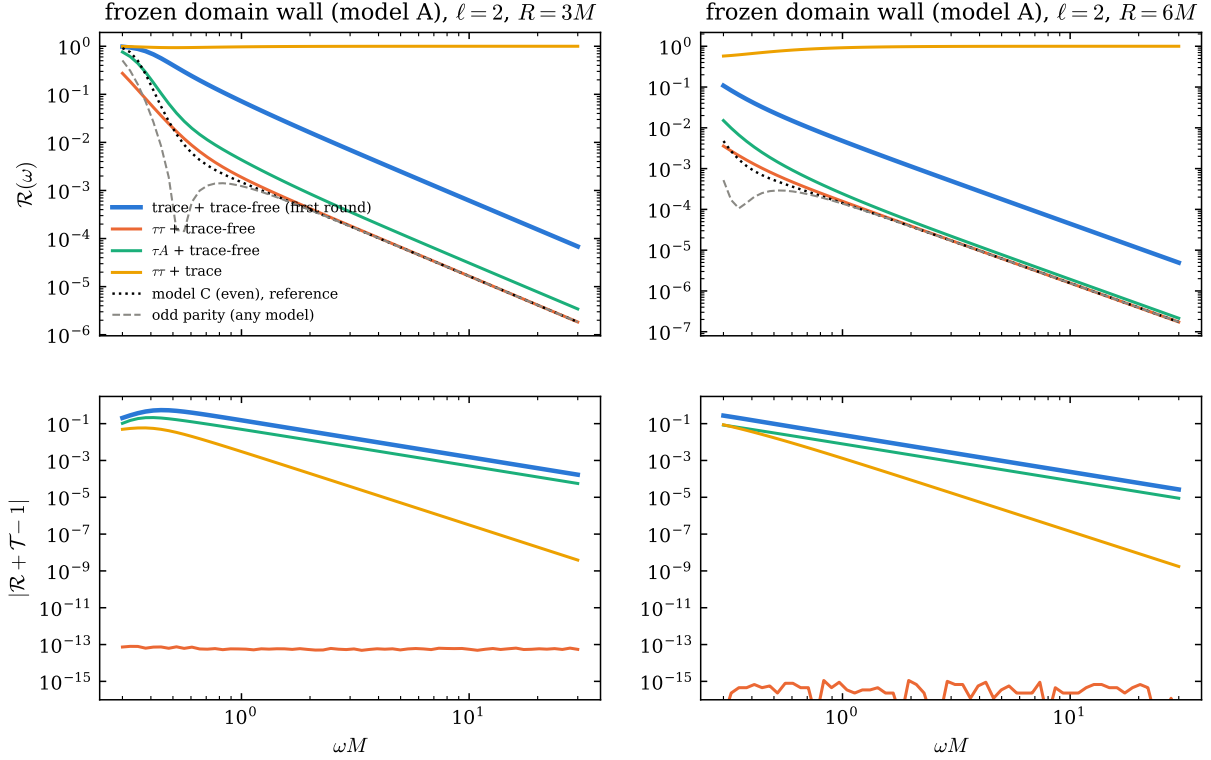


Figure 4: Reflectivity and energy balance of the frozen domain wall for four admissible prescriptions (second round). The first-round prescription is the thick blue curve.

$M \rightarrow 0$. It also regenerated the symbolic matrices from the script and found them identical. The poles lie at $\omega(R^3/M)^{1/2} \approx 1.54$ and $0.985i$ ($R = 6M$): on the matter scale, but not at the matter QNMs themselves.

5.4 Model A: an adiabatic matching and its limits

For an exact solution of the moving-shell problem, the $\tau\tau$ and τA Israel equations follow from the others through the Codazzi identity and $D_b S^{ab} = 0$. That is why the overdetermined odd system of model A is consistent. In the even sector the frozen, monochromatic ansatz is not an exact solution at the turning point: $\dot{T}_0 = \beta/f$ has $\ddot{T}_0 = 0$ but $\dddot{T}_0 \propto \ddot{R}^2 \neq 0$. Correspondingly $\text{rank } \mathbf{A} = 6$ while $\text{rank}[\mathbf{A}|\mathbf{B}] = 8$. The first round imposed the four metric-continuity equations and the two angular Israel equations (called “evolution” equations) and monitored the $\tau\tau$ and τA rows (“constraints”). The constraint violation then decreases as ω^{-1} , and $|\mathcal{R} + \mathcal{T} - 1|$ decreases as ω^{-2} (Figs. 8 and 6).

Second-round verification. For a frozen ansatz that is not a solution, the choice of which two of the four Israel equations to impose is a *prescription*. The second round kept metric continuity and tried all six choices (Fig. 4). The resulting reflectivities differ at $O(1)$ even at $\omega M = 30$, from total reflection to the δ -barrier law, and the lowest QNMs differ by 20–30%. With the first round’s choice the energy error is *larger than $\mathcal{R}$ itself* at all high frequencies. The choice “ $\tau\tau$ + trace-free” conserves energy exactly and reproduces the universal high-frequency law of the odd sector and of model C. The even-parity results of model A are therefore *not* an $O(\omega_{\text{dyn}}/\omega)$ -accurate approximation. They should be regarded as undetermined by the frozen-background idealization. The odd sector of model A is unaffected.

6 The scattering problem

6.1 Definitions

The interior is regular and nothing is absorbed, so the physical (closed) system reflects everything. Outside, $\Psi_+ \rightarrow B_{\text{in}} e^{-i\omega r_*} + B_{\text{out}} e^{+i\omega r_*}$ with $|B_{\text{out}}/B_{\text{in}}| = 1$ for real ω , so $S = B_{\text{out}}/B_{\text{in}} = e^{2i\delta(\omega)}$, and the QNMs appear as resonances of the phase δ . The same is true for the inviscid star of Boyanov et al. [13], whose $\mathcal{R}^2 = 1$ in that limit. To obtain non-trivial coefficients we treat the *exterior potential plus shell* as a scatterer between infinity and an “open” interior that contains only the ingoing wave $\Psi_- = \hat{h}_\ell^{\text{in}}(\nu r)$. We then define

$$\mathcal{R}(\omega) = \left| \frac{B_{\text{out}}}{B_{\text{in}}} \right|^2, \quad \mathcal{T}(\omega) = \frac{1}{|B_{\text{in}}|^2}. \quad (6.1)$$

The Wronskian flux $J = \text{Im}(\Psi^* \partial_x \Psi)$ is constant on each side: $J_{\text{int}} = \sqrt{f_R} \nu \text{Im}(\hat{h}^{\text{in}*} \hat{h}^{\text{in}'}) = -\omega$ and $J_{\text{ext}} = \omega(|B_{\text{out}}|^2 - |B_{\text{in}}|^2)$. If the junction conserves J , then $\mathcal{R} + \mathcal{T} = 1$. For a real $\mathbb{T}$ this is the statement $\det \mathbb{T} = f_R^{-1/2}$. This definition is a physically motivated convention: it describes waves that, once through the shell, never return. It is the analogue of the horizon boundary condition of a black hole, and it makes the black-hole limit transparent.

6.2 Limits

- $\omega \rightarrow 0$: $\mathcal{R} \rightarrow 1$. The irregular static branch $r^{-\ell}$ of the interior Hankel function dominates, and the matched exterior solution is a standing wave (the shell’s tidal response). Long waves do not see the shell, as for the star of Boyanov et al.
- $R \rightarrow 2M$: $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$. The shell sits deep in the near-horizon region, where $V \rightarrow 0$ and $\Delta_{\text{odd}} \rightarrow 0$, so the transmitted wave is exactly the horizon-ingoing one of Vishveshwara’s problem [3].
- $M \rightarrow 0$: $\mathcal{R} \rightarrow 0$ (flat space).
- High frequency. The potentials are negligible and only the δ barrier remains. With $\Psi = e^{-i\omega x} + r e^{i\omega x}$ outside and $t e^{-i\omega x}$ inside, continuity and $[\Psi'] = \Delta \Psi$ give $r = \Delta/(2i\omega - \Delta)$, so

$$\mathcal{R} \simeq \frac{\Delta_{\text{odd}}^2}{4\omega^2 + \Delta_{\text{odd}}^2}. \quad (6.2)$$

No impedance mismatch arises, because the local wavenumber $\omega/\sqrt{f_R}$ in proper distance is continuous across the shell. The thin shell becomes *transparent* at high frequency, unlike the viscous star of Boyanov et al., which absorbs.

7 Numerical methods

7.1 Method 1: transfer matrix with a continued-fraction exterior basis

The purely outgoing exterior solution is expanded about a point $r = R_2$ [11, 17]:

$$\Psi_{\text{up}} = \chi(r) \sum_{n \geq 0} a_n z^n, \quad z = 1 - \frac{R_2}{r}, \quad \chi = (r - 2M)^{2iM\omega} e^{i\omega r}. \quad (7.1)$$

The identity $f\chi'/\chi = i\omega$ turns Eq. (3.2) into $c(z)\phi'' + d(z)\phi' + e(z)\phi = 0$ with $u = 1 - z$ and $m = M/R_2$:

$$c = u^2(1 - 2mu), \quad d = -2u + 6mu^2 + 2i\omega R_2, \quad e = 6mu - \lambda. \quad (7.2)$$

This is Pani et al.’s (B4) with $\omega \rightarrow -\omega$ (they use $e^{+i\omega t}$). Writing $c = \sum c_j z^j$ etc., the coefficients obey the four-term recurrence $\alpha_n a_{n+1} + \beta_n a_n + \gamma_n a_{n-1} + \delta_n a_{n-2} = 0$, with $\alpha_n = n(n+1)c_0$,

$\beta_n = n(n-1)c_1 + nd_0$, $\gamma_n = (n-1)(n-2)c_2 + (n-1)d_1 + e_0$, $\delta_n = (n-2)(n-3)c_3 + (n-2)d_2 + e_1$. Gaussian elimination reduces it to three terms [16] (Pani’s B10–B13). The minimal solution [18] is then selected by the continued fraction $a_1/a_0 = -\hat{\gamma}_1/(\hat{\beta}_1 - \hat{\alpha}_1\hat{\gamma}_2/(\hat{\beta}_2 - \dots))$ (B17), whose depth is doubled until it converges to 10^{-12} . Then $\Psi'_{\text{up}}/\Psi_{\text{up}}|_{R_2} = i\omega R_2/(R_2 - 2M) + (a_1/a_0)/R_2$. The series must converge for $0 \leq z < 1$, so the singular point $r = 2M$, at $z = 1 - R_2/2M$, must lie outside the unit disk: $R_2 \geq 4M$ (Pani’s “ $R_2 > 2$ ” in units $2M = 1$). The code uses $R_2 = \max(R, 4.5M)$ and integrates from R_2 down to R when needed. Zerilli solutions follow from Eq. (3.9). For real ω , $\Psi_{\text{in}} = \Psi_{\text{up}}^*$, and the normalization $|\Psi_{\text{up}}(R)|^2 = \omega/\text{Im}(\partial_{r_*} \ln \Psi_{\text{up}})$ follows from the Wronskian.

7.2 Method 2: shooting

The exterior equation is integrated outward in r from R^+ to $r_{\text{far}} = \max(150M, 60/\omega)$ with DOP853 [47, 48] (`rtol` = 10^{-10} , `atol` = 10^{-12}). At r_{far} the solution is projected on $\Psi_{\text{up}} = e^{i\omega r_*} \sum_k c_k r^{-k}$ and Ψ_{up}^* , whose coefficients follow from $f u'' + (f' + 2i\omega)u' - (\lambda/r^2 - 6M/r^3)u = 0$:

$$2i\omega(k+1)c_{k+1} = [k(k+1) - \lambda]c_k - 2M(k^2 - 4)c_{k-1}, \quad c_0 = 1. \quad (7.3)$$

The series is asymptotic and is truncated at its smallest term. The reflectivity estimator of Boyanov et al. (App. B1), $|i\omega\Psi + \partial_{r_*}\Psi|/|i\omega\Psi - \partial_{r_*}\Psi| = |B_{\text{out}}/B_{\text{in}}|$ averaged over four periods, is also evaluated. The two methods share only the junction matrix and the interior functions.

7.3 Quasinormal modes

QNMs are the zeros of the normalized Wronskian at R^+ between the solution coming from the regular interior through the junction, $\mathbf{y}_J = \mathbf{T}(\omega)(\hat{j}_\ell, \nu\hat{j}'_\ell)$, and the outgoing solution: $W = (\Psi_J\Psi'_{\text{up}} - \Psi'_J\Psi_{\text{up}})/(|\mathbf{y}_J||\mathbf{y}_{\text{up}}|)$. Method 1 takes Ψ_{up} from the continued fraction. Method 2 follows the inward strategy of Boyanov et al.: it starts from the asymptotic series at a far point on the complex ray $r = R + se^{i\varphi}$, $\varphi = \pi/2 - \arg \omega$, along which Ψ_{up} is exponentially subdominant, and integrates back to R . Candidates come from scans of $\log |W|$ in the complex plane, with extra scans near the real axis for compact shells and post-Newtonian seeds (PSP26 Eq. 7.8) for the matter modes. They are refined with `scipy.optimize.root`, and every root is re-converged with Method 2.

7.4 Schwarzschild black hole and WKB

Black-hole QNMs come from Leaver’s continued fraction [15] (units $2M = 1$, $\rho = -i\omega$, $\alpha_n = n^2 + (2\rho+2)n + 2\rho+1$, $\beta_n = -[2n^2 + (8\rho+2)n + 8\rho^2 + 4\rho + \ell(\ell+1) - 3]$, $\gamma_n = n^2 + 4\rho n + 4\rho^2 - 4$), with inversion index equal to the overtone number. The black-hole reflectivity follows from integration from $r = 2M(1 + 10^{-6})$ with a Frobenius start. The WKB QNM formula of arbitrary order is obtained from the equivalent perturbative (Bender–Wu [27]) expansion of an inverted anharmonic oscillator [24], computed with symbolic overtone n through the hypervirial–Hellmann–Feynman recursion [25, 26],

$$2kE\langle x^{k-1} \rangle = 2k\langle x^{k-1}V \rangle + \langle x^k V' \rangle - \frac{1}{2}k(k-1)(k-2)\langle x^{k-3} \rangle, \quad (7.4)$$

with the r_* -derivatives of V at its maximum. Truncation after $g^{2(N-1)}$ reproduces the N -th order formula: Schutz–Will [20] for $N = 1$, Iyer–Will [21, 22] for $N = 3$, Konoplya [23] for $N = 6$. For modes trapped between the flat interior and the exterior barrier of ultracompact shells, the WKB estimate is a Bohr–Sommerfeld condition with Langer’s replacement $\lambda \rightarrow (\ell + \frac{1}{2})^2$ [28],

$$\int_{r_1}^R \sqrt{\nu^2 - \frac{(\ell+1/2)^2}{r^2}} dr + \int_R^{r_a} \sqrt{\omega^2 - V} \frac{dr}{f} = \pi(n + \frac{1}{2}), \quad (7.5)$$

plus Gamow tunnelling, $\text{Im} \omega = -T_b/(4\omega \int dx/\sqrt{\omega^2 - V})$ with $T_b = \exp(-2 \int_{r_a}^{r_b} \sqrt{V - \omega^2} dr/f)$.

8 Results

8.1 Validation

Table 2: Validation summary of the first round (M1/M2: transfer-matrix and shooting methods; $|\Delta_{12}|$: maximum difference over $\omega M \in [0.005, 2]$).

Check	value
Odd: generated vs analytic $\mathcal{T}$	5.7e-16
Odd: $\mathcal{T}^A - \mathcal{T}^C$	1.1e-15
Even C: $ \mathcal{T} - \mathbb{I} $ at $M = 10^{-8}$ (transparent limit)	4.0e-09
Even A: $ \mathcal{T} - \mathbb{I} $ at $M = 10^{-8}$	4.7e-09
Even C: $\max \det \mathcal{T} \sqrt{f_R} - 1 $	3.6e-12
Even A: constraint violation, $R = 3M$, $\omega M = 1$	1.2e-01
Even A: constraint violation, $R = 3M$, $\omega M = 10$	1.4e-02
Even A: constraint violation, $R = 3M$, $\omega M = 100$	1.4e-03
Even A: constraint violation, $R = 6M$, $\omega M = 1$	6.0e-02
Even A: constraint violation, $R = 6M$, $\omega M = 10$	7.9e-03
Even A: constraint violation, $R = 6M$, $\omega M = 100$	7.9e-04
odd, $\ell = 2$, $R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.3e-10 / 5.3e-10 / 2.5e-06
odd, $\ell = 2$, $R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	6.0e-10 / 6.0e-10 / 2.0e-06
even-C, $\ell = 2$, $R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.3e-10 / 5.3e-10 / 4.0e-06
even-C, $\ell = 2$, $R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	6.1e-10 / 6.1e-10 / 3.2e-06
even-A, $\ell = 2$, $R = 3M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	5.4e-01 / 6.0e-10 / 2.5e-06
even-A, $\ell = 2$, $R = 6M$: $\max_{\omega} \mathcal{R} + \mathcal{T} - 1 $ (M2) / $\max \Delta_{12} $ / Boyanov estimator	2.2e+00 / 6.1e-10 / 3.6e-06
Low frequency: $\max 1 - \mathcal{R} $ at $\omega M = 10^{-3}$ (all models)	8.9e-16

Energy conservation. For odd parity and for model C, $|\mathcal{R} + \mathcal{T} - 1| \lesssim 6 \times 10^{-10}$ over $\omega M \in [0.005, 2]$ (Fig. 8), with $\mathcal{R}$ and $\mathcal{T}$ obtained independently from the asymptotic amplitudes. For model A the violation is an error of the frozen approximation and decays as ω^{-2} .

Consistency of the methods. Transfer matrix and shooting agree to $\lesssim 10^{-9}$ on $\mathcal{R}$ and $\mathcal{T}$ (Fig. 7). On the QNMs they agree to 10^{-10} – 10^{-13} for most modes, and to better than 6.4×10^{-5} for all 397 catalogued modes; the task required 10^{-4} .

Pure Schwarzschild. The exterior machinery gives $\omega M = 0.3736716844 - 0.0889623157i$ for $\ell = 2$, $n = 0$. All overtones $n \leq 3$ for $\ell = 2, 3, 4$ agree with Berti, Cardoso and Starinets’ tables [19] (Table 3). The black-hole reflectivity is identical for RW and Zerilli to 10^{-7} (isospectrality) and reproduces Vishveshwara’s Fig. 2: $\mathcal{R} = 1.55 \times 10^{-2}$ at $(2M\omega)^2 = 1$ and 5.3×10^{-9} at $(2M\omega)^2 = 4$.

WKB. Relative errors for $\ell = 2$, $n = 0$ are 6.6×10^{-2} , 1.5×10^{-3} and 2.3×10^{-4} at orders 1, 3 and 6 (Fig. 5). They decrease with ℓ and grow with n . The third-order values coincide with the explicit Iyer–Will formula; e.g. $\ell = 2$: $0.3732 - 0.0892i$ ($n = 0$) and $0.3460 - 0.2749i$ ($n = 1$).

8.2 Scattering coefficients

Figures 9–11 show $\mathcal{R}(\omega)$ and $\mathcal{T}(\omega)$.

1. At low frequency the shell is a perfect reflector. For compact shells ($R \lesssim 3M$) the transition to transparency happens at $\omega \sim \sqrt{V_{\max}}$, where the exterior barrier dominates. For dilute shells it happens at $\omega R \sim \ell$, where the flat-space centrifugal barrier is crossed.
2. For compact shells the curves approach the Schwarzschild reflectivity. Even and odd curves differ: the shell spacetime breaks the isospectrality of Schwarzschild, as PSP26 also note for the QNMs.

Table 3: Schwarzschild QNMs ωM : Leaver continued fraction vs WKB of orders 1, 3, 6.

ℓ	n	Leaver CF (this code)	WKB 1st	WKB 3rd	WKB 6th
2	0	$0.37367168 - 0.08896232i$	$0.3988 - 0.0883i$	$0.37316 - 0.08922i$	$0.373619 - 0.088891i$
2	1	$0.34671100 - 0.27391488i$	$0.4534 - 0.2330i$	$0.34602 - 0.27492i$	$0.346297 - 0.273480i$
2	2	$0.30105345 - 0.47827698i$	$0.5170 - 0.3406i$	$0.30293 - 0.47106i$	$0.298520 - 0.477560i$
2	3	$0.25150496 - 0.70514820i$	$0.5775 - 0.4268i$	$0.24746 - 0.67290i$	$0.241715 - 0.709595i$
3	0	$0.59944329 - 0.09270305i$	$0.6166 - 0.0923i$	$0.59927 - 0.09273i$	$0.599443 - 0.092703i$
3	1	$0.58264380 - 0.28129811i$	$0.6619 - 0.2580i$	$0.58235 - 0.28141i$	$0.582642 - 0.281290i$
3	2	$0.55168490 - 0.47909275i$	$0.7251 - 0.3925i$	$0.55320 - 0.47668i$	$0.551594 - 0.479047i$
3	3	$0.51196191 - 0.69033710i$	$0.7909 - 0.5038i$	$0.51575 - 0.67743i$	$0.511099 - 0.690468i$
4	0	$0.80917838 - 0.09416396i$	$0.8223 - 0.0939i$	$0.80910 - 0.09417i$	$0.809178 - 0.094164i$
4	1	$0.79663153 - 0.28433435i$	$0.8602 - 0.2694i$	$0.79650 - 0.28437i$	$0.796631 - 0.284334i$
4	2	$0.77270953 - 0.47990818i$	$0.9187 - 0.4204i$	$0.77364 - 0.47897i$	$0.772695 - 0.479900i$
4	3	$0.73983673 - 0.68392432i$	$0.9844 - 0.5492i$	$0.74331 - 0.67830i$	$0.739665 - 0.683902i$

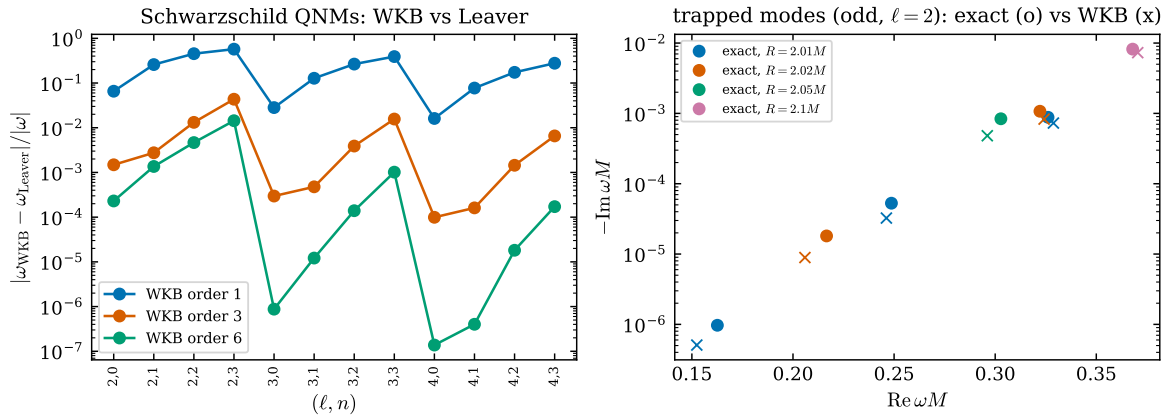


Figure 5: WKB validation (first round). Left: relative error of WKB orders 1, 3, 6 vs Leaver. Right: trapped modes of ultracompact shells, exact (o) vs Bohr–Sommerfeld/Gamow WKB (x).

3. At high frequency $\mathcal{R} \rightarrow 0$ following the δ -barrier law (6.2) (checked to 10^{-3} at $\omega M = 8$): the thin shell is transparent.
4. Even parity, model C, shows narrow resonant features on the matter scale $\omega \sim (M/R^3)^{1/2}$. They depend on Γ (Fig. 11); elsewhere the Γ dependence is weak.
5. Model C (even) merges with the odd-parity curve at high frequency: the fluid’s inertia and pressure response become irrelevant ($\mathcal{R}_{\text{even}}/\mathcal{R}_{\text{odd}} = 1.0000$ at $\omega M = 10$).

Second-round verification. The first round also reported that the frozen domain wall (model A) “reflects much more strongly” at high frequency (green curves of Figs. 9, 6). The second round found this to be an artifact of the model-A prescription (Sec. 5.4, Fig. 4), so that statement is not retained here.

8.3 Quasinormal modes

The spectra are shown in the complex plane (Fig. 13) and as functions of R/M (Fig. 14).

- *Wave modes* exist in both sectors, with $\omega \sim 1/R$. For dilute shells PSP26 derived the Newtonian law $\text{Re}(\omega R) \simeq n\pi$ or $(n + \frac{1}{2})\pi$ and $\text{Im}(\omega R) \simeq -\xi$, with $\xi = \frac{1}{2} \ln[2\ell(\ell + 1)R/M]$ (their Eqs. 7.23–7.25). The interior cavity leaks through the weakly reflecting shell, so the damping grows only logarithmically with R/M . Our wave modes coincide with PSP26’s Figs. 3–4 to the reading precision (Table 5).

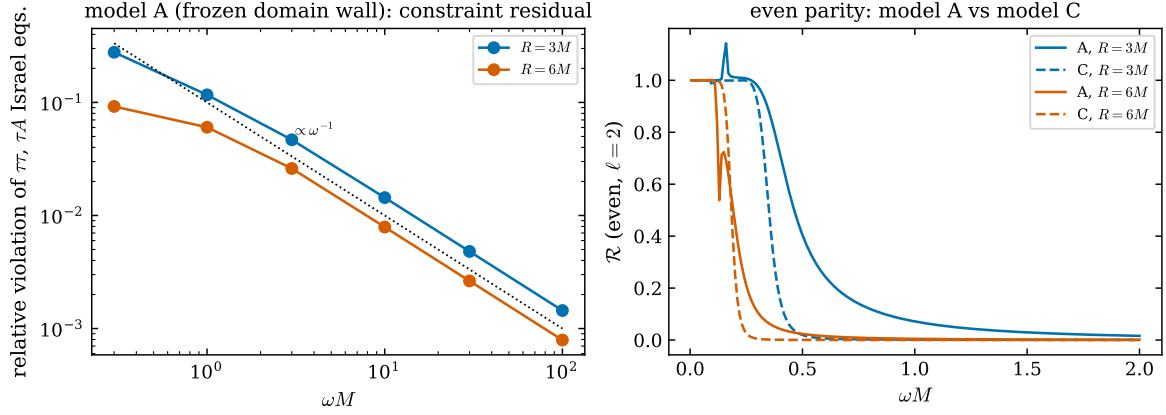


Figure 6: Model A with the first-round prescription. Left: relative violation of the $\tau\tau$ and τA Israel equations, $\propto \omega^{-1}$. Right: even-parity reflectivity of models A and C (but see Fig. 4).

Table 4: Approach to the black-hole reflectivity ($\ell = 2$; even: model C, 40 digits).

ωM	BH	odd 2.2M	odd 2.05M	odd 2.002M	even-C 2.2M	even-C 2.05M	even-C 2.01M
0.20	0.99913	0.99886	0.99916	0.99907	0.99966	0.99921	0.99915
0.30	0.94521	0.92247	0.93661	0.94729	0.96577	0.95037	0.94996
0.37	0.56378	0.46896	0.52601	0.56105	0.62593	0.61107	0.54639
0.45	0.07559	0.04894	0.10167	0.07539	0.13163	0.07064	0.07361
0.60	0.00066	0.00554	0.00319	0.00021	0.00814	0.00166	0.00141

- For compact shells ($R < 3M$) a family of *trapped* modes appears below the barrier top, damped only by tunnelling. As $R \rightarrow 2M$ they become exponentially long-lived and dense (Table 7). These are the modes responsible for gravitational-wave “echoes” [41, 43]. None of them tends to a Schwarzschild QNM: the black-hole spectrum is *not* a limit of the shell spectrum, although the reflectivity is.
- *Matter modes* exist only in the even sector of model C (Sec. 8.4).

Table 5: Wave modes ($\ell = 2$; even: model C, $\Gamma = 2$) compared with PSP26 Figs. 3–4 (reading uncertainty ± 0.02).

parity	M/R	mode	$\text{Re}(R\omega)$	$\text{Im}(R\omega)/\xi$	PSP26 Re	PSP26 Im/ ξ
odd	0.167	1	1.484	-0.632	1.47	-0.630
odd	0.167	2	4.151	-0.835	4.10	-0.840
even	0.250	1	1.033	-0.651	1.05	-0.655
even	0.100	1	0.615	-0.825	0.62	-0.825
even	0.100	2	2.978	-0.907	2.95	-0.910

8.4 Matter modes and the instability of the static shell

PSP26 found four even-parity matter modes per ℓ : a purely imaginary unstable mode, its damped mirror, and a pair $\pm \text{Re}\omega + i \text{Im}\omega$ with small negative imaginary part. They concluded that the shell is dynamically unstable on all angular scales. Their method is different from ours. They use retarded-time slicing, an RW-type master function in both sectors, and Chebyshev collocation or Mano–Suzuki–Takasugi (MST) type expansions. Reproducing their results is therefore an independent test of our even junction (Table 8, Fig. 15). At $M/R = 0.2$, $\Gamma = 2$, $\ell = 2$ we find $\omega(R^3/M)^{1/2} = 0.60771i$ for the unstable mode and $1.26707 - 0.00110i$ for the stable pair, as in

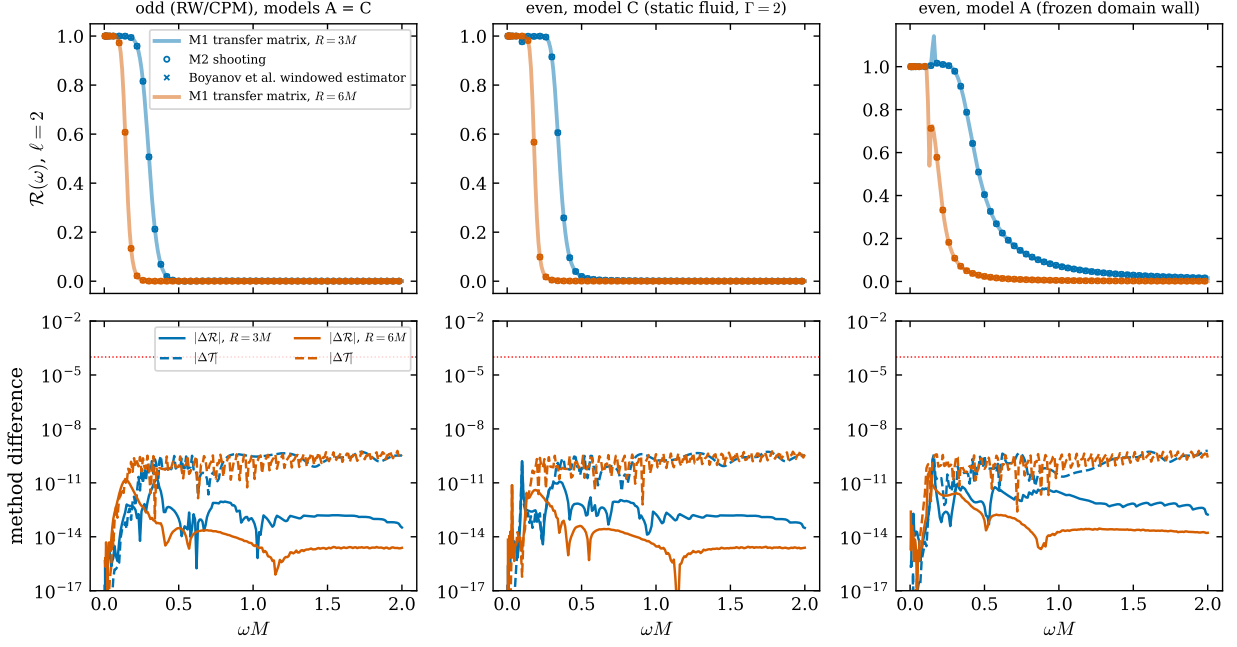


Figure 7: Transfer matrix (lines) vs shooting (circles). The bottom row shows the absolute differences; the red line is the required 10^{-4} .

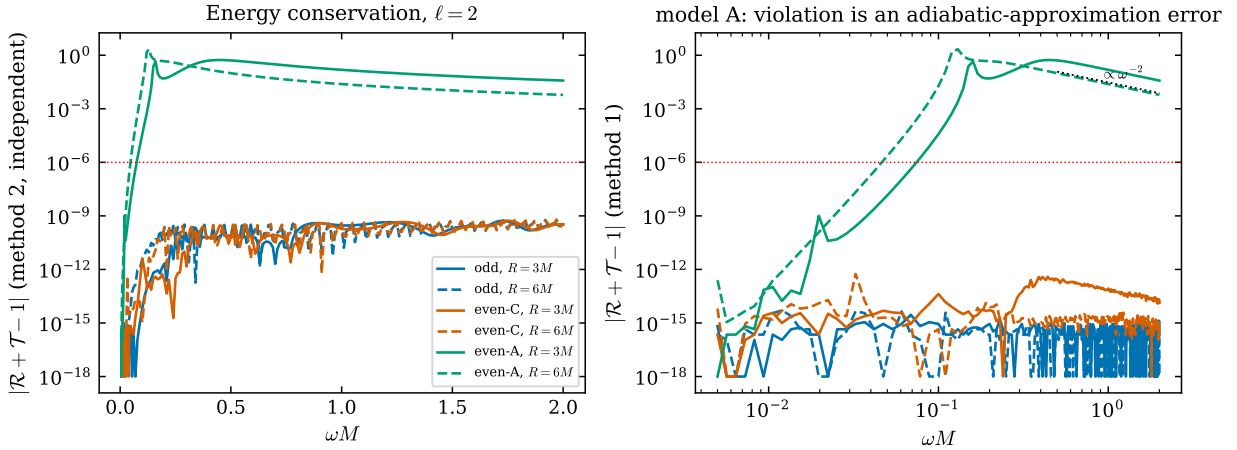


Figure 8: Energy conservation $|\mathcal{R} + \mathcal{T} - 1|$. Left: Method 2. Right: Method 1 on log-log axes (model A decays as ω^{-2}).

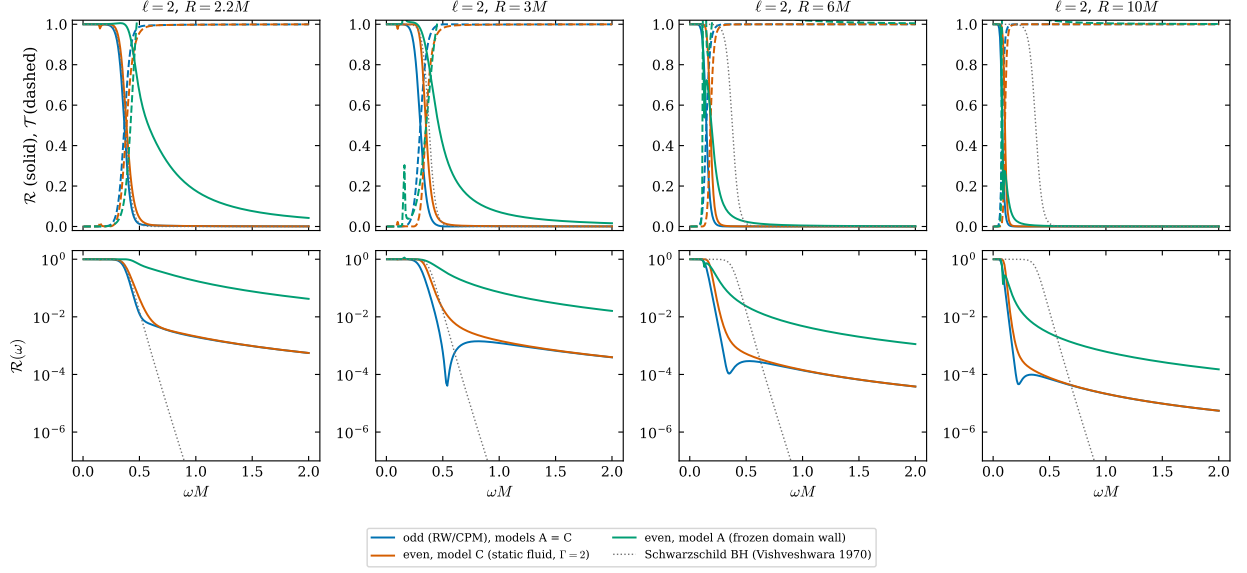


Figure 9: $\mathcal{R}$ (solid) and $\mathcal{T}$ (dashed; top) and $\mathcal{R}$ on a log scale (bottom), $\ell = 2$, $R/M = 2.2, 3, 6, 10$. Blue: odd (any model). Orange: even, model C ($\Gamma = 2$). Green: even, model A (first-round prescription; see Sec. 5.4). Grey: Schwarzschild black hole.

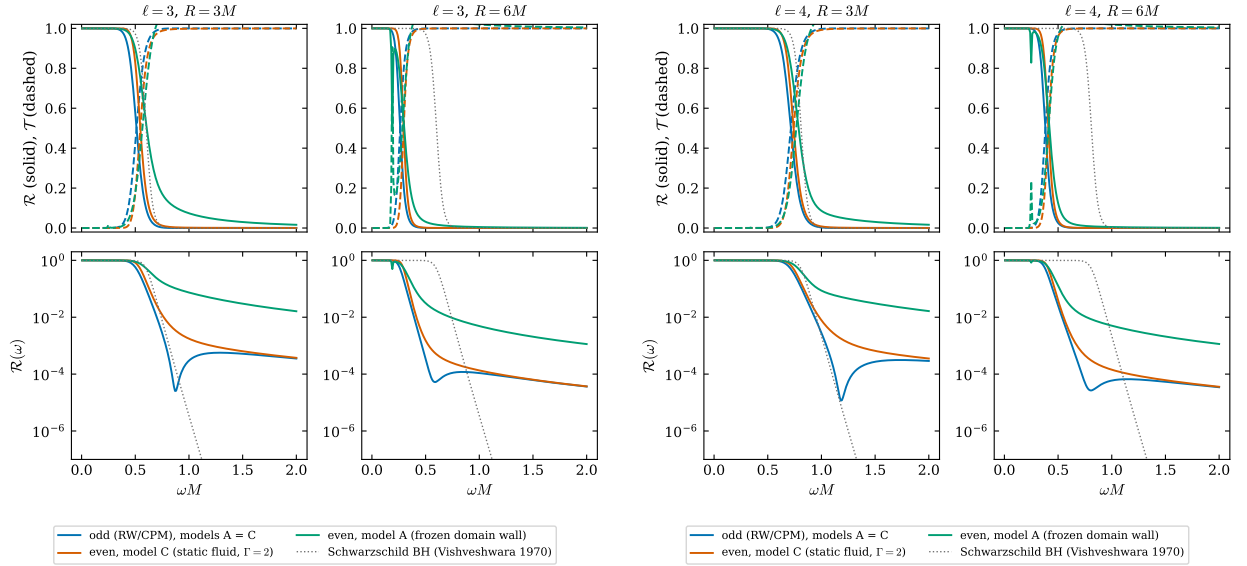


Figure 10: As Fig. 9 for $\ell = 3$ (left) and $\ell = 4$ (right), $R/M = 3, 6$.

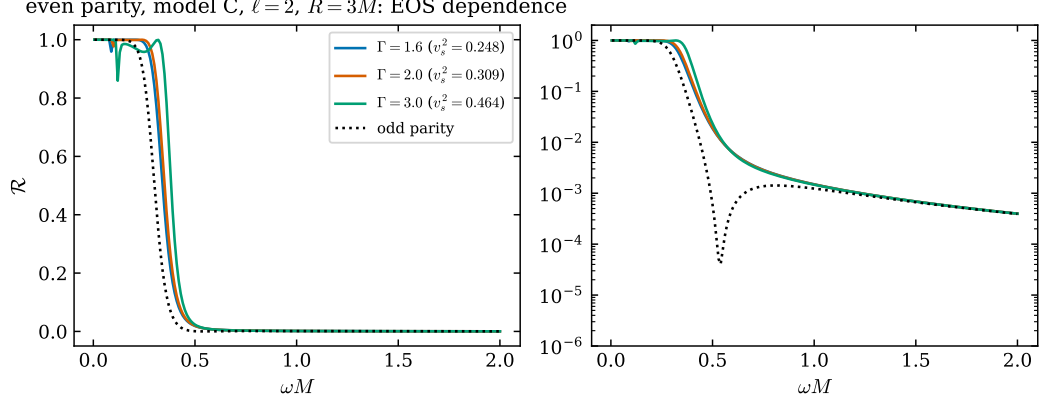


Figure 11: Dependence of the even-parity reflectivity (model C, $\ell = 2$, $R = 3M$) on the adiabatic index.

Table 6: Lowest wave modes, $\ell = 2$, from the first-round catalogue. The stable matter pair is missing at $R = 2.2M$ and $3M$ (see the box in Sec. 8.4).

R/M	sector	lowest wave modes ωM (ordered by $\text{Re } \omega$; Methods 1 and 2 agree to $< 10^{-4}$)
2.2	odd	$0.42111 - 0.03502i$, $0.66196 - 0.19736i$, $0.66692 - 0.28043i$, $0.78377 - 0.71437i$
2.2	even-C	$0.44694 - 0.04385i$, $0.57996 - 0.19970i$, $0.69756 - 0.56050i$, $0.92055 - 0.21412i$; matter: $0.00000 + 0.24429i$, $0.00000 - 0.24429i$
2.2	even-A	$0.10284 - 0.04070i$, $0.45445 - 0.10428i$, $0.56024 - 0.44388i$, $0.65595 - 0.85878i$
3	odd	$0.20371 - 1.01489i$, $0.35352 - 0.55904i$, $0.42638 - 0.17728i$, $0.97955 - 0.33508i$
3	even-C	$0.21102 - 0.76599i$, $0.40766 - 0.29425i$, $0.61594 - 0.31862i$, $1.91546 - 0.38053i$; matter: $-0.00000 + 0.13321i$, $-0.00000 - 0.13326i$
3	even-A	$0.05288 - 0.97817i$, $0.24913 - 0.59916i$, $0.35675 - 0.22995i$, $0.96224 - 0.13441i$
6	odd	$0.24739 - 0.22534i$, $0.69189 - 0.29749i$, $1.13482 - 0.31819i$, $1.57166 - 0.33553i$
6	even-C	$0.13710 - 0.26885i$, $0.45011 - 0.28793i$, $0.91288 - 0.30736i$, $1.35361 - 0.32702i$; matter: $0.08942 - 0.00005i$, $-0.00000 + 0.04014i$, $-0.00000 - 0.04014i$
6	even-A	$0.18120 - 0.22393i$, $0.66401 - 0.17021i$, $1.12337 - 0.19838i$, $1.56453 - 0.21844i$
10	odd	$0.15342 - 0.18490i$, $0.45405 - 0.22153i$, $0.74518 - 0.23457i$, $1.03239 - 0.24567i$
10	even-C	$0.06152 - 0.19748i$, $0.29784 - 0.21714i$, $0.59930 - 0.22783i$, $0.88902 - 0.24022i$; matter: $0.04421 - 0.00001i$, $-0.00000 + 0.01762i$, $0.00000 - 0.01762i$
10	even-A	$0.11383 - 0.17403i$, $0.43293 - 0.14212i$, $0.73603 - 0.15791i$, $1.02659 - 0.17041i$

PSP26 Figs. 1–2. As $M/R \rightarrow 0$ the values approach the post-Newtonian limits $0.51470i$ and 1.50496 of PSP26 Eq. (7.8), with $O(M/R)$ differences. The static model C is therefore linearly unstable, with growth time $\simeq 1.6(R^3/M)^{1/2}$.

Second-round verification. Newtonian check. The second round derived the non-radial oscillations of a Newtonian self-gravitating fluid shell from scratch (App. A) and recovered PSP26’s Eq. (7.8) identically, for all ℓ and Γ . This also exposed a misprint in PSP26’s quartic (7.6); its closed-form solution (7.8), used here, is correct. The Γ dependence (PSP26 Figs. 1–2, right panels) is reproduced as well. **Catalogue.** The first-round catalogue misses the stable matter pair at $R = 2.2, 2.5, 3M$ ($\ell = 2$), e.g. $0.197121 - 0.000310i$ at $3M$, and the unstable $\ell = 4$ mode at $R = 2.2M$ ($0.43726i$). These omissions do not change the conclusion: an unstable mode exists for every ℓ and R examined.

9 Discussion

The shell model. Neither a pure domain wall nor a dust shell has a static configuration; a static shell needs positive surface pressure, $p/\sigma = (1 - \sqrt{f_R})/4\sqrt{f_R}$. Even that shell is dynamically unstable (PSP26, confirmed). The “collapse problem” of the prior notes is thus generic, but its time scales differ. The fluid shell’s instability grows on $\sim 1.6(R^3/M)^{1/2}$, which is long compared with the wave periods, so frequency-domain scattering off model C is a meaningful adiabatic idealization

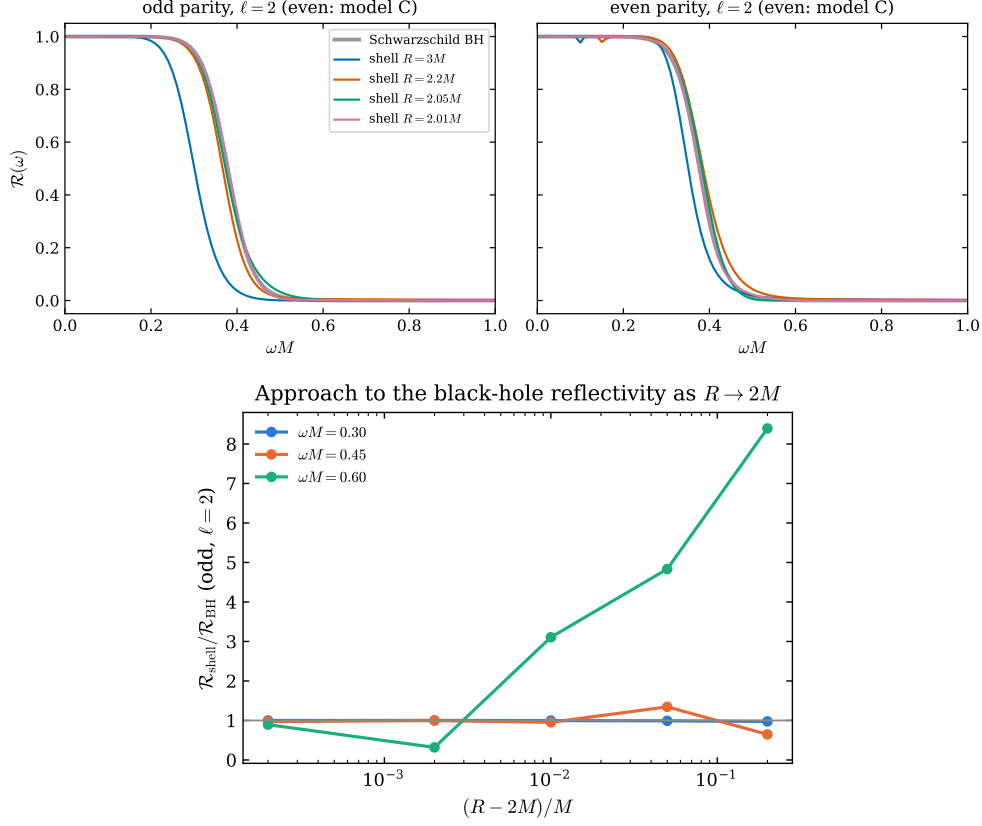


Figure 12: Top: approach of the shell reflectivity to the black-hole value as $R \rightarrow 2M$ (first round; see Table 4). Bottom (second round): the approach is oscillatory and slow, with the shell contribution scaling like $\Delta_{\text{odd}} \propto (R - 2M)^{1/2}$.

for $\omega(R^3/M)^{1/2} \gg 1$. The domain wall collapses on $\sim R/\sqrt{2}$, comparable to the wave periods, and its frozen even-parity problem is not a controlled approximation (Sec. 5.4).

Junction conditions. In the odd sector, $[\Psi] = 0$ and $[\partial_x \Psi] = \Delta_{\text{odd}} \Psi$ hold with $\Delta_{\text{odd}} = \sqrt{f_R}(1 - \sqrt{f_R})/R$, independent of ω and of the equation of state. The $[K]^{(0)}$ term that worried the prior notes cancels against $\delta S_{\tau A}$. The even sector gives a frequency-dependent 2×2 matrix, because the shell's radial displacement differs between the two RW gauges and its compressional dynamics enter through v_s^2 .

Relation to previous work. The odd junction agrees with PSP26 and with Pani et al.; the even junction reproduces PSP26's matter and wave modes. Pani et al.'s gravastar has $\Sigma = 0$ and a de Sitter interior, so only the structure of the matching conditions can be compared; PSP26 argue that unstable modes should also exist in that model. The new elements are:

- the reflection and transmission coefficients with their low-frequency, high-frequency and black-hole limits;
- the explicit δ -function form of the odd junction (and its thick-shell derivation);
- a critical assessment of the frozen domain-wall model;
- two independent numerical methods, and an independent third code in the verification round.

Limitations. Model C is restricted to $R \geq 25M/12$ by the energy and causality conditions. No time-domain evolution of the unstable shell was attempted. The open-interior definition of $\mathcal{R}$ and

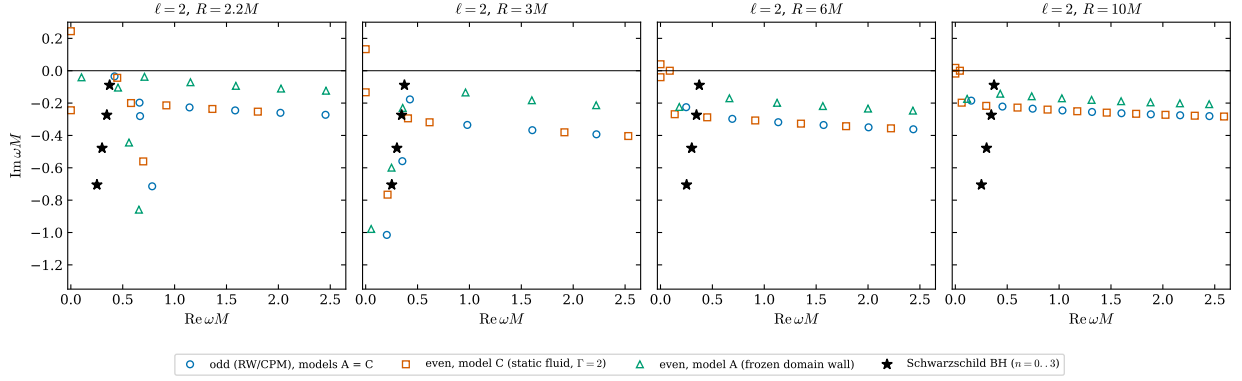


Figure 13: QNMs for $\ell = 2$: odd (circles), even model C (squares), even model A (triangles; first-round prescription) and Schwarzschild (stars). The unstable matter mode lies on the positive imaginary axis.

Table 7: Trapped modes of ultracompact shells (odd, $\ell = 2$): exact vs WKB.

R/M	exact (M1)	WKB (Bohr–Sommerfeld + Gamow)
2.01	$0.162603 - 9.711e - 07 i$	$0.152432 - 5.081e - 07 i$
2.01	$0.248683 - 5.284e - 05 i$	$0.246194 - 3.249e - 05 i$
2.01	$0.326133 - 8.798e - 04 i$	$0.328792 - 7.287e - 04 i$
2.02	$0.216610 - 1.809e - 05 i$	$0.205905 - 8.922e - 06 i$
2.02	$0.322125 - 1.072e - 03 i$	$0.324183 - 8.289e - 04 i$
2.05	$0.302815 - 8.384e - 04 i$	$0.296095 - 4.811e - 04 i$
2.10	$0.367989 - 8.195e - 03 i$	$0.370495 - 7.304e - 03 i$

$\mathcal{T}$ is a convention adapted to the question “how much of an incident wave is sent back by the barrier and the shell”; for the closed physical system $|S| = 1$.

10 Conclusions

A spherical thin shell with a flat interior is the simplest horizonless compact object, and its perturbation theory can be carried out in closed form up to a 2×2 matching problem. The odd parity reduces to a δ -potential, the even parity to a transfer matrix that conserves flux exactly for the static fluid shell. Scattering interpolates between perfect reflection at low frequency, the Schwarzschild reflectivity for $R \rightarrow 2M$, and transparency at high frequency. The QNM spectrum contains wave, trapped and matter modes, and the growing matter mode shows that the only static member of the family is unstable. All of this was reproduced and cross-checked independently. The one exception is the even sector of the frozen domain wall, which the frozen approximation does not determine.

A Newtonian oscillations of a self-gravitating fluid shell

Consider a shell of radius R , surface density σ , two-dimensional pressure p and mass $M = 4\pi R^2 \sigma$ ($G = 1$). The gravitational field acting on the shell is the average of the interior (zero) and exterior ($-M/R^2$) fields. The normal force of a 2D fluid with pressure p on a surface of mean curvature $2H$ is $2Hp$, outward. Equilibrium therefore requires $2p/R = \sigma M/2R^2$, i.e. $p = \sigma M/4R$, the Newtonian limit of κ in Eq. (2.7). Let the Lagrangian displacement be $\xi_r Y \hat{r} + \xi_h \nabla_{S^2} Y$, with time dependence $e^{-i\omega t}$, and define $x = \xi_r/R$ and $y = \xi_h/R$. Then:

- continuity: $\Delta\sigma/\sigma = -(2x - \lambda y)$, and $\Delta p = \Gamma p \Delta\sigma/\sigma$;

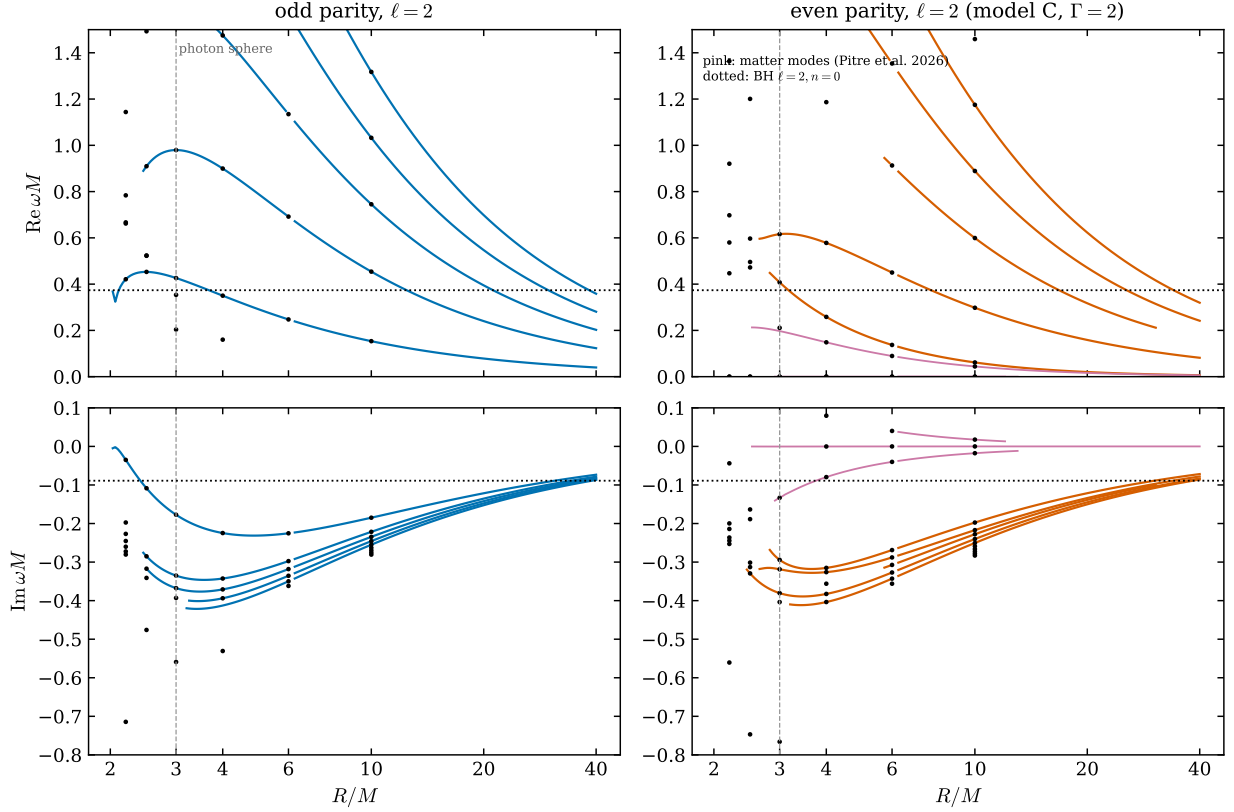


Figure 14: QNM frequencies vs R/M for $\ell = 2$ (continuation from $R = 6M$; dots: independent scans).

- self-gravity: the displaced sheet acts on the exterior like a surface density $\sigma(\lambda y + \ell x)Y$ and on the interior like $\sigma(\lambda y - (\ell + 1)x)Y$ (the radial displacement shifts the multipole moments);
- mean curvature: $2H = 2/R - (2 - \lambda)xY/R$.

With $\varsigma^2 = \omega^2 R^3/M$, the radial and tangential equations of motion become

$$\varsigma^2 x = -x + \frac{\lambda(y + 2x)}{2(2\ell + 1)} + \frac{\Gamma}{2}(2x - \lambda y) + \frac{(2 - \lambda)x}{4} - \frac{2x - \lambda y}{2}, \quad (\text{A.1})$$

$$\varsigma^2 y = -\frac{\Gamma}{4}(2x - \lambda y) - \frac{\lambda y - (\ell + 1)x}{2\ell + 1}. \quad (\text{A.2})$$

The trace and determinant of this 2×2 problem are

$$\varsigma_+^2 + \varsigma_-^2 = \frac{(\ell^2 + \ell + 4)\Gamma - (\ell^2 + \ell + 6)}{4}, \quad \varsigma_+^2 \varsigma_-^2 = -\frac{(\ell - 1)\ell(\ell + 1)[(\ell + 2)(2\ell - 3)\Gamma - 4(\ell - 2)]}{16(2\ell + 1)}. \quad (\text{A.3})$$

The roots coincide identically with PSP26's Eq. (7.8). Because the product is negative for $\Gamma > 4(\ell - 2)/[(\ell + 2)(2\ell - 3)]$, one ς^2 is negative, i.e. there is a purely imaginary growing mode: the Newtonian origin of the instability. For $\ell = 2$ and $\Gamma = 2$ the roots are $\varsigma = 1.504962$ and $0.514695 i$. PSP26's Eq. (7.6) prints the constant term as $-(\ell - 1)\ell(\ell + 1)(\ell + 2)[(2\ell - 3)\Gamma - 4]/[16(2\ell + 1)]$. That form disagrees with (7.8) and with their own Newtonian Eq. (10.62), and for $\ell = 2$ it would predict stability.

Table 8: Even-parity matter modes, model C, $\ell = 2$, $\Gamma = 2$ ($|\Delta_{12}|$: M1–M2 difference).

M/R	unstable: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $	stable pair: $\omega(R^3/M)^{1/2}$	$ \Delta_{12} $
0.30	0.668693 i	1.6e-12	1.095903 – 1.70e – 03 i	1.8e-12
0.25	0.636634 i	7.1e-12	1.188088 – 1.51e – 03 i	9.2e-13
0.20	0.607709 i	1.6e-11	1.267067 – 1.10e – 03 i	3.9e-13
0.15	0.581398 i	2.3e-11	1.336362 – 6.47e – 04 i	5.1e-13
0.10	0.557313 i	3.3e-11	1.398197 – 2.78e – 04 i	6.1e-13
0.05	0.535158 i	4.6e-11	1.454057 – 5.79e – 05 i	7.4e-13
0.02	0.522690 i	4.6e-11	1.485144 – 6.52e – 06 i	8.3e-13
0.01	0.518662 i	6.1e-11	1.495140 – 1.20e – 06 i	0.0e+00
$\rightarrow 0$ (PN, PSP26 Eq. 7.8)	0.514695 i		1.504962	

B Axial perturbations of an anisotropic fluid and the thin-shell limit

For $ds^2 = -e^{2\Phi}dt^2 + e^{2\Lambda}dr^2 + r^2d\Omega^2$, with $e^{-2\Lambda} = 1 - 2m/r$ and $T_{\mu\nu} = (\rho + p_t)u_\mu u_\nu + p_t g_{\mu\nu} + (p_r - p_t)s_\mu s_\nu$, consider an axial perturbation h_0, h_1 (RW gauge) and δu_ϕ . The $\theta\phi$ Einstein equation gives h_0 in terms of h_1, h'_1 . The $r\phi$ equation then becomes, for $X = e^{\Phi-\Lambda}h_1/r$ and $dr_* = e^{\Lambda-\Phi}dr$,

$$X_{,r_*r_*} + \{\omega^2 - e^{2\Phi}[\ell(\ell+1)/r^2 - 6m/r^3 + 4\pi(\rho - p_r)]\}X = 0, \quad (\text{B.1})$$

once $m' = 4\pi r^2\rho$, $\Phi' = (m + 4\pi r^3 p_r)/r(r - 2m)$ and the anisotropic TOV equation $p'_r = -(\rho + p_r)\Phi' + 2(p_t - p_r)/r$ are used. For $p_r = 0$ the TOV equation gives $p_t = \rho r\Phi'/2$. Integrating across a layer whose thickness tends to zero, one finds $\int \rho d\ell_p = (1 - \sqrt{f_R})/4\pi R = \sigma$ and $\int p_t d\ell_p = (1 - \sqrt{f_R})^2/16\pi R\sqrt{f_R} = p$, exactly the model-C background. The only singular term of the potential is $4\pi\rho e^{2\Phi} \rightarrow \Delta_{\text{odd}}\delta(x - x_R)$ with $\Delta_{\text{odd}} = 4\pi\sigma\sqrt{f_R}$. Since $X \propto \Psi_{\text{CPM}}$ on both sides, this is the junction (5.5).

C Code map and reproducibility

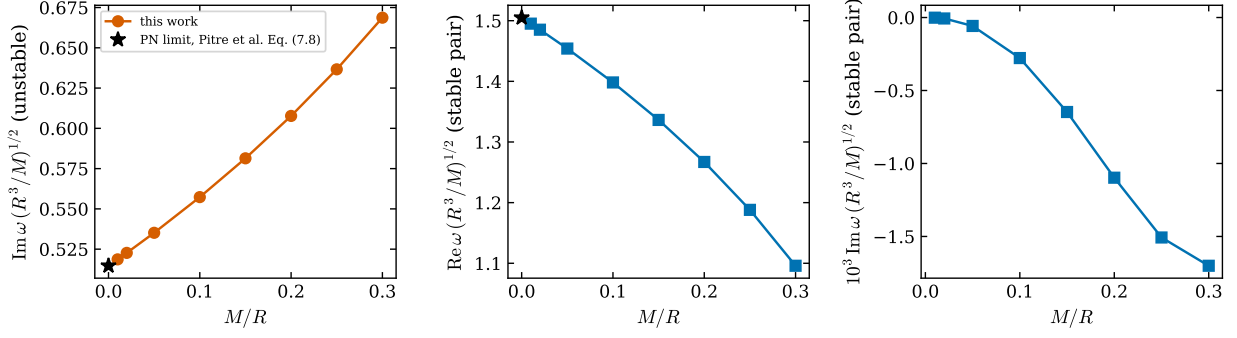
First round/code/main.py	driver: validate, scatter, qnm, tracks, csv, figures
symbolic/vacuum_relations.py	Einstein-tensor linearization; Eqs. (3.3), (3.6)–(3.8)
symbolic/derive_junction.py	junction conditions; exports A, B
symbolic/exterior_series.py	continued-fraction ODE; Chandrasekhar map
shellgw/*.py	background, interior, exterior, junction, scattering, QNM, black hole, WKB
Second round/checks/*.py	independent verification codes (companion report)

Re-running the first round reproduces `validation.json` bit for bit, all tables and all figures (pixel-identical) and the symbolic junction matrices.

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Even-parity matter modes of the static fluid shell (model C), $\ell = 2$, $\Gamma = 2$ -- compare Pitre, Schneider & Poisson (2026) Figs. 1, 2



Even-parity matter modes of the static fluid shell (model C, $\ell = 2$)

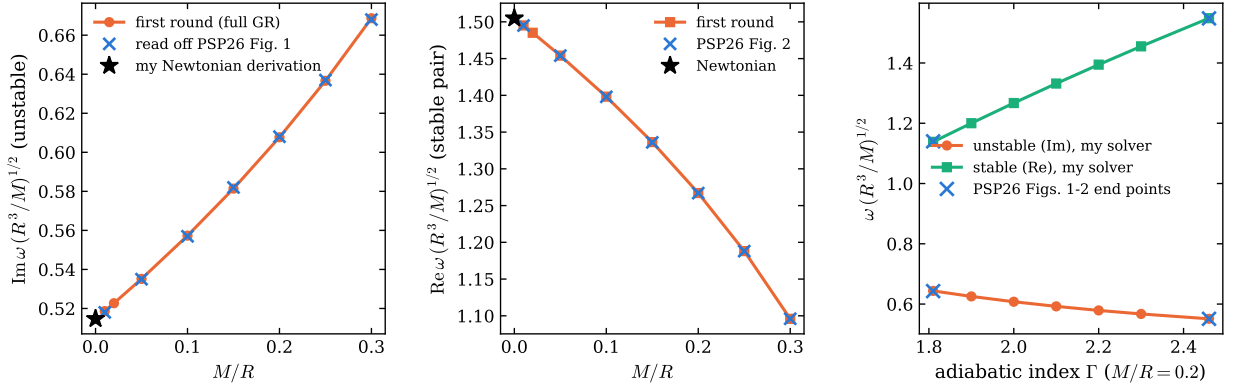


Figure 15: Top: matter modes vs compactness with the post-Newtonian limits (first round). Bottom (second round): comparison with values read off PSP26, our Newtonian limit, and the Γ dependence at $M/R = 0.2$.

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